Hybrid Zeroth- and First-Order Method for Stochastic Nonsmooth Minimax Optimization

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Abstract

Minimax optimization problems are central to many machine learning applications. Prior work has largely focused on smooth minimax objectives or composite problems with a single nonsmooth component. In contrast, general nonsmooth minimax optimization has received significantly less attention, and existing methods remain limited in both theoretical guarantees and practical performance. In this work, we investigate a class of nonsmooth minimax problems and propose a hybrid optimization algorithm that integrates zeroth-order and first-order information. We establish a non-asymptotic convergence guarantee and show that, by incorporating momentum methods, the proposed method achieves a stochastic oracle complexity of $\tilde{\mathcal{O}}(d_1^{1.5}\delta^{-1}\epsilon^{-4})$ for finding a (δ,ϵ)-generalized Goldstein stationary point (GGSP), where d_1 is the dimension of the first variable, δ is the perturbation radius in zeroth-order gradient. We further demonstrate the empirical effectiveness of our method on AUC maximization and robust classification, where it consistently outperforms competitive baselines.

1 Introduction

Minimax optimization plays a central role in classical game theory and underlies a broad spectrum of emerging machine learning applications. Representative examples include robust classification [7, 38, 26], federated learning [41, 30, 34], adversarial training [8], and few-shot meta-learning [36, 15, 45]. As modern machine learning problems continue to scale in dimensionality and exhibit increasing nonconvexity and real-world uncertainty, the demand for efficient minimax optimization algorithms with rigorous theoretical guarantees has become increasingly pressing.

The most natural and widely adopted approach for solving minimax problems is gradient descent-ascent (GDA), a generalization of gradient descent that updates the two players either simultaneously or in an alternating manner. Several representative methods are summarized in Table 1. For example, [18] established complexity guarantees for simultaneous GDA and stochastic GDA (Stoc-GDA) in finding stationary points. They showed that when the objective is strongly concave in the second variable, GDA requires $\mathcal{O}(\epsilon^{-2})$ iterations, while Stoc-GDA requires $\mathcal{O}(\epsilon^{-4})$ samples with a batch size of $\mathcal{O}(\epsilon^{-2})$ to reach an ϵ-stationary point. [28] proposed a multi-step GDA scheme for deterministic minimax problems and proved convergence to a stationary point with complexity $\mathcal{O}(\log(\epsilon^{-1})\epsilon^{-3.5})$. [13] combined adaptive learning rates with variance reduction and showed that their method achieves an ϵ-stationary point with complexity $\mathcal{O}(\epsilon^{-3})$. In addition, zeroth-order methods have also been explored for minimax optimization [36, 12].

Despite these advances, most existing theoretical guarantees rely critically on smoothness assumptions and therefore do not readily extend to nonsmooth minimax problems. Motivated by this limitation, recent works have begun to investigate nonsmooth settings, with a primary focus on composite minimax formulations, where the objective consists of a smooth term together with nonsmooth but

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Table 1: Comparison of representative methods for minimax optimization. In Column 4, "1st" denotes first-order methods, "0th" denotes zeroth-order methods, and "Subgrad." denotes subgradient-based methods. In Column 5, **C**, **NC**, **SC**, and **WC** stand for concave, nonconvex, strongly concave, and weakly concave objectives, respectively. Column 6 reports the complexity of finding an ϵ-stationary point or its corresponding variants, and $\tilde{\mathcal{O}}(\cdot)$ is used to hide logarithmic factors (Here, for the first-order and subgradient methods, we use the first-order oracle complexity, and for our method and the zeroth-order methods, we use the zeroth-order oracle complexity.).

Problem	Method	Ref.	Order	Objective	Complexity
Smooth	GDA	[18]	1st	NC-SC	$\mathcal{O}(\epsilon^{-2})$
	Stoc-GDA	[18]	1st	NC-SC	$\mathcal{O}(\epsilon^{-4})$
	Multi-Step GDA	[28]	1st	NC-SC	$\mathcal{O}(\log(\epsilon^{-1})\epsilon^{-3.5})$
	AdaGDA	[13]	1st	NC-SC	$\mathcal{O}(\epsilon^{-4})$
	VR-AdaGDA	[13]	1st	NC-SC	$\mathcal{O}(\epsilon^{-3})$
	ZO-GDMSA	[39]	0th	NC-SC	$\mathcal{O}((d_1+d_2)\epsilon^{-2})$
	ZO-SGDMSA	[39]	0th	NC-SC	$\mathcal{O}((d_1+d_2)\epsilon^{-4})$
	ZO-SGDA	[39]	0th	NC-SC	$\mathcal{O}((d_1+d_2)\epsilon^{-4})$
	Acc-ZOMDA	[12]	0th	NC-SC	$\mathcal{O}((d_1+d_2)\epsilon^{-3})$
Composite	Smoothed-PLDA	[17]	1st	NC-SC	$\mathcal{O}(\epsilon^{-4})$
	SAMDAM	[48]	1st	NC-SC	$\mathcal{O}(\epsilon^{-4})$
Nonsmooth	TTSGDA	[19]	1st	WC-SC	$\tilde{\mathcal{O}}(\epsilon^{-6})$
	TTSGDA	[19]	1st	WC-C	$\mathcal{O}(\epsilon^{-8})$
	ZOFOSGDA	Ours	0th+1st	NC-SC	$\tilde{\mathcal{O}}(d_1^{1.5}\delta^{-1}\epsilon^{-4})$

structured components such as regularizers or constraints. For example, [48] proposed a stochastic alternating mirror descent-ascent method with momentum and established a sample complexity of $\mathcal{O}(\epsilon^{-4})$ for finding an ϵ-stationary point. [17] developed a proximal linear descent–ascent scheme and proved convergence to a stationary point with complexity $\mathcal{O}(\epsilon^{-4})$. While these methods effectively exploit composite structures, they typically rely on decomposability assumptions that may not hold for fully general nonsmooth minimax problems.

To go beyond the composite setting, [19] proposed a two-timescale GDA method for nonsmooth weakly-convex-strongly-concave minimax optimization based on subgradient updates and established an $\mathcal{O}(\epsilon^{-6})$ complexity bound for reaching an ϵ-stationary point. However, two-timescale approaches generally require a stringent separation between step sizes (e.g., $\eta_x/\eta_y \to 0$), along with additional structural conditions, which complicate algorithm design and parameter tuning. Moreover, the resulting convergence complexity is relatively high, leading to slow practical convergence and limiting the applicability of such methods in large-scale nonsmooth settings. Consequently, designing efficient algorithms for general nonsmooth minimax optimization remains a challenging open problem.

To address these challenges, we develop a novel algorithmic framework for general nonsmooth minimax optimization that does not assume weak convexity. Inspired by the notion of (δ,ϵ)-Goldstein stationary points for unconstrained nonconvex nonsmooth optimization [47, 20, 1, 31, 4, 16], we introduce a new (δ,ϵ)-stationarity concept tailored to nonsmooth minimax problems. Building on this refined notion of approximate stationarity, we propose a hybrid zeroth- and first-order method with momentum. We rigorously show that the proposed method finds a (δ,ϵ)-generalized Goldstein stationary point (GGSP) using $\tilde{\mathcal{O}}(d_1^{1.5}\delta^{-1}\epsilon^{-4})$ zeroth-order oracle calls, where d_1 denotes the dimension of the first variable. Finally, we validate the effectiveness of our approach through extensive experiments on AUC maximization and robust classification, where the proposed algorithm consistently outperforms existing baselines.

Our contributions can be summarized as follows.

1. We propose ZOFOSGDA, the first zeroth-order algorithm specifically designed for *nonsmooth* nonconvex–strongly concave minimax problems—a highly general and challenging setting that has remained unexplored in prior zeroth-order methods. By integrating momentum for both primal and dual variables, ZOFOSGDA effectively navigates the complex landscape of nonsmooth objectives, leading to substantially more stable and efficient optimization.

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2. We introduce a (δ, ϵ)-generalized Goldstein stationary point—a new, practically meaningful optimality notion for nonsmooth minimax problems. We prove that ZOFOSGDA converges to such a point with a zeroth-order oracle complexity of $\tilde{\mathcal{O}}(d_1^{1.5}\delta^{-1}\epsilon^{-4})$, delivering, to our knowledge, the first non-asymptotic convergence guarantee in this broad nonsmooth setting.

3. Extensive empirical evaluations on two representative applications demonstrate that ZOFOSGDA not only consistently outperforms existing methods but also exhibits remarkable robustness to nonsmooth components. This provides strong evidence for the method's practical effectiveness in real-world scenarios where nonsmoothness is inevitable, and highlights the value of our general theoretical framework

2 Related Works

In this section, we give a brief review of existing nonsmooth nonconvex optimization and nonconvex minimax optimization.

2.1 Nonsmooth Optimization

In this subsection, we review related work on classical nonsmooth nonconvex optimization. Recently, the development of non-asymptotic convergence analyses for nonsmooth optimization has attracted significant attention. [47] established non-asymptotic oracle complexity bounds for interpolated normalized gradient descent to reach a (δ, ϵ)-GGSP of a Lipschitz objective, assuming access to a nonstandard subgradient oracle. [6] and [35] subsequently refined this approach by introducing random perturbations at each iteration, thereby eliminating the need for such oracle assumptions. [18] proposed zeroth-order methods, referred to as GFM, for nonsmooth optimization by establishing a connection between the Goldstein δ-subdifferential and randomized smoothing. They showed that the proposed method converges to a (δ, ϵ)-GGSP with an oracle complexity of $\mathcal{O}(d^{1.5}\delta^{-1}\epsilon^{-4})$. Subsequent works further improved these results. For example, [1] and [21] leveraged variance-reduction techniques and proved that their methods can find a (δ, ϵ)-GGSP with improved complexity $\mathcal{O}(d^{1.5}\delta^{-1}\epsilon^{-3})$. Kornowski and Shamir [16] obtained a sharper bound by applying the reduction technique introduced in [4] to the zeroth-order setting. Moreover,[22] and [23] extended these methodologies to constrained and compositional settings, respectively. However, these approaches are not directly applicable to nonsmooth nonconvex minimax optimization problems.

2.2 Minimax Optimization

Minimax optimization has been widely applied in areas such as adversarial robustness and multi-agent decision-making. In recent years, numerous first-order methods have been developed for solving smooth minimax problems [28, 18, 48, 42, 43, 44]. Building on these works, several accelerated gradient descent–ascent algorithms have been proposed for stochastic minimax optimization by leveraging variance-reduction techniques [29, 24, 13]. In addition, composite nonconvex–strongly-concave minimax optimization has been studied in [14, 2]. Huang and Gao [11] further proposed a class of Riemannian gradient descent–ascent algorithms for geodesically nonconvex–strongly-concave minimax problems on Riemannian manifolds. Zeroth-order methods have also been investigated for minimax optimization. Despite their strong theoretical guarantees, most existing approaches rely critically on smoothness assumptions. As a result, nonsmooth minimax optimization remains relatively underexplored. To address this gap, [19] proposed a two-timescale gradient descent–ascent (GDA) algorithm for nonsmooth weakly-convex–strongly-concave minimax problems and established a convergence rate of $\mathcal{O}(\epsilon^{-6})$. However, this complexity is relatively high, which may lead to slow convergence and limit the practicality of such methods in large-scale nonsmooth settings. Therefore, designing more efficient algorithms for general nonsmooth minimax optimization remains an open and challenging problem.

3 Preliminaries

In this section, we introduce the background for nonsmooth minimax optimization and randomized smoothing technique in zeroth-order optimization.

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3.1 Notations

Throughout the paper, $\|\cdot\|$ denotes the ℓ_2 norm for vectors and the spectral norm for matrices. I_d denotes the d-dimensional identity matrix, and $\mathbf{1}_d$ denotes the d-dimensional vector with all entries equal to one. Given a function $f(\mathbf{x}, \mathbf{y})$, we use $f(\mathbf{x}, \cdot)$ to denote the function with respect to the second variable while keeping $\mathbf{x}$ fixed, and $f(\cdot, \mathbf{y})$ to denote the function with respect to the first variable while keeping $\mathbf{y}$ fixed. The Euclidean ball centered at $\mathbf{x}$ with radius δ is defined as $\mathbb{B}_\delta(\mathbf{x}) := \{\mathbf{y} \in \mathbb{R}^d : \|\mathbf{y} - \mathbf{x}\| \leq \delta\}$. We use $\mathrm{Conv}(A)$ to denote the convex hull of note the set A. When reporting complexity results, the notation $\tilde{\mathcal{O}}(\cdot)$ is used to hide logarithmic factors.

3.2 Problem Setting

In this paper, we study the following stochastic nonsmooth minimax optimization problem:

$$\min_{\mathbf{x} \in \mathbb{R}^{d_1}} \max_{\mathbf{y} \in \mathcal{Y} \subseteq \mathbb{R}^{d_2}} f(\mathbf{x}, \mathbf{y}) := \mathbb{E}_\xi[f(\mathbf{x}, \mathbf{y}; \xi)], \tag{1}$$

where $f(\cdot, \mathbf{y})$ is possibly nonconvex and nonsmooth, $f(\mathbf{x}, \cdot)$ is differentiable and strongly concave, ξ is a random variable following an unknown distribution, and $\mathcal{Y} \subseteq \mathbb{R}^{d_2}$ is a closed convex set.

Throughout the paper, we adopt the following standard assumptions, which are commonly used in the literature on minimax optimization [17, 18, 14, 12, 19].

Assumption 3.1. The objective function $f(\mathbf{x}, \mathbf{y})$ and the constraint set $\mathcal{Y}$ satisfy the following conditions:

1. $f(\mathbf{x}, \mathbf{y})$ is almost everywhere differentiable with respect to $\mathbf{x}$ and continuously differentiable with respect to $\mathbf{y}$.
2. For any fixed $\mathbf{y} \in \mathcal{Y}$, the function $f(\cdot, \mathbf{y})$ may be nonconvex, and for any fixed $\mathbf{x} \in \mathbb{R}^{d_1}$, the function $f(\mathbf{x}, \cdot)$ is τ-strongly concave.
3. The function $f(\mathbf{x}, \mathbf{y})$ is L-Lipschitz continuous in $(\mathbf{x}, \mathbf{y})$ for some constant $L > 0$.
4. The gradient $\nabla_y f(\mathbf{x}, \mathbf{y})$ is L_{fy}-Lipschitz continuous in $(\mathbf{x}, \mathbf{y})$.

3.3 Background for Nonsmooth Analysis

We begin by recalling the notion of generalized gradients for nondifferentiable functions [3].

Definition 3.2. Given a point $\mathbf{x} \in \mathbb{R}^d$ and a direction $\mathbf{v} \in \mathbb{R}^d$, the generalized directional derivative of a nondifferentiable function f is given by $f^\circ(\mathbf{x}; \mathbf{v}) := \limsup_{\mathbf{y} \to \mathbf{x}, t \downarrow 0} \frac{f(\mathbf{y}+t\mathbf{v}) - f(\mathbf{y})}{t}$. Then, the generalized gradient of f is defined as a set $\partial f(\mathbf{x}) := \{\mathbf{h} \in \mathbb{R}^d : \mathbf{h}^\top \mathbf{v} \leq f^\circ(\mathbf{x}; \mathbf{v}), \forall \mathbf{v} \in \mathbb{R}^d\}$.

Based on the Clarke subdifferential, an ϵ-Clarke stationary point $\mathbf{x}$ of the nonsmooth optimization problem $\min_{\mathbf{x}} f(\mathbf{x})$ is defined as a point satisfying

$$\min\{\|\mathbf{h}\| : \mathbf{h} \in \partial f(\mathbf{x})\} \leq \epsilon. \tag{2}$$

Since directly computing ϵ-Clarke stationary points of a nonsmooth minimization problem within finite time is generally intractable, Zhang *et al.* [47] and Lin *et al.* [20] introduced a refined notion of approximate stationarity based on the Goldstein δ-subdifferential.

Definition 3.3. [3]. Given a Lipschitz function $f : \mathbb{R}^d \to \mathbb{R}$, a point $\mathbf{x}$ and $\delta > 0$, the Goldstein δ-subdifferential of f at $\mathbf{x}$ is defined as $\partial_\delta f(\mathbf{x}) := \mathrm{Conv}(\bigcup_{\mathbf{y} \in \mathbb{B}_\delta(\mathbf{x})} \partial f(\mathbf{y}))$, which is the convex combination of the generalized gradients at point in the δ-neighbourhood of $\mathbf{x}$.

The relationship between the Goldstein δ-subdifferential and the Clarke subdifferential is summarized as follows [20, 22, 1].

Lemma 3.4. *The Goldstein δ-subdifferential is equivalent to the Clarke subdifferentiable when $\delta = 0$, i.e., $\partial_0 f(\mathbf{x}) = \partial f(\mathbf{x})$.*

Based on the Goldstein δ-subdifferential, a refined notion of approximate stationarity for nonsmooth nonconvex optimization is defined as follows [47, 20].

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Definition 3.5. Given a Lipschitz function $f : \mathbb{R}^d \to \mathbb{R}$, a point $\mathbf{x}$ and $\delta \geq 0$, a point $\mathbf{x}$ is called a (δ, ϵ)-generalized Goldstein stationary point of f if $\min\{\|\mathbf{h}\| : \mathbf{h} \in \partial_\delta f(\mathbf{x})\} \leq \epsilon$.

Recently, a series of algorithms [47, 20, 1, 22] have been developed with non-asymptotic convergence guarantees for finding (δ, ϵ)-generalized Goldstein stationary points with $\delta > 0$.

4 Zeroth- and First-Order Stochastic Gradient Descent-Ascent

In this section, we first introduce notions of approximate stationarity for nonsmooth, nonconvex-strongly-concave minimax optimization and establish several fundamental properties. We then present an efficient algorithmic framework for solving such problems.

4.1 Definitions of Approximate Stationarity

Equipped with the notions of generalized gradients and Goldstein subdifferentials, we are now ready to define stationary points for nonsmooth minimax optimization problems.

Definition 4.1. A pair $(\mathbf{x}, \mathbf{y}) \in \mathbb{R}^{d_1} \times \mathcal{Y}$ is called a (δ, ϵ)-generalized Goldstein stationary point (GGSP) if it satisfies

$$\min\{\|\mathbf{h}\| : \mathbf{h} \in \partial_{\delta\mathbf{x}} f(\mathbf{x}, \mathbf{y})\} \leq \epsilon, \ \left\|\mathcal{P}_{\mathcal{Y}}\left(\mathbf{y} + \tfrac{1}{L_{fy}}\nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y})\right) - \mathbf{y}\right\| \leq \epsilon / L_{fy}, \tag{3}$$

where $\partial_{\delta\mathbf{x}} f(\mathbf{x}, \mathbf{y})$ denotes the Goldstein δ-subdifferential of $f(\mathbf{x}, \mathbf{y})$ with respect to $\mathbf{x}$.

By setting $\delta = 0$, a pair $(\mathbf{x}, \mathbf{y}) \in \mathbb{R}^{d_1} \times \mathcal{Y}$ is said to be an ϵ-generalized Clarke stationary point (GCSP) of Problem (1) if it satisfies

$$\min\{\|\mathbf{h}\| : \mathbf{h} \in \partial_{\mathbf{x}} f(\mathbf{x}, \mathbf{y})\} \leq \epsilon, \ \left\|\mathcal{P}_{\mathcal{Y}}\left(\mathbf{y} + \tfrac{1}{L_{fy}}\nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y})\right) - \mathbf{y}\right\| \leq \epsilon / L_{fy}, \tag{4}$$

where $\partial_{\mathbf{x}} f(\mathbf{x}, \mathbf{y})$ denotes the generalized gradient of f with respect to $\mathbf{x}$, and $\partial_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}) = \partial_{0\mathbf{x}} f(\mathbf{x}, \mathbf{y})$.

We adopt the notion of (δ, ϵ)-GGSP instead of ϵ-GCSP primarily due to considerations of computational efficiency. Recent work by [19] demonstrates that, for nonsmooth minimax optimization, a variant of ϵ-GCSP can be attained via a two-timescale gradient descent-ascent scheme; however, the resulting complexity is $\mathcal{O}(\epsilon^{-6})$, which is relatively high. By allowing $\delta > 0$, the ϵ-GGSP notion relaxes exact Clarke stationarity and leads to a more tractable stationarity criterion, enabling algorithms with substantially improved non-asymptotic complexity guarantees. Importantly, ϵ-GGSP reduces to ϵ-GCSP when $\delta = 0$, and the two notions coincide in the smooth setting as shown in [18], thereby preserving consistency with classical stationarity concepts. This relaxation is therefore essential for the design of provably efficient algorithms for general nonsmooth minimax problems.

4.2 Randomized Smoothing

Randomized smoothing offers a straightforward yet effective technique that naturally extends to minimax optimization. Formally, given an L-Lipschitz function $f(\mathbf{x}, \mathbf{y})$, which may be nonsmooth and nonconvex, and a random vector $\mathbf{u}$ sampled from a distribution $\mathbb{P}$, we define the smoothed objective as

$$f_\delta(\mathbf{x}, \mathbf{y}) := \mathbb{E}_{\mathbf{u}}[f(\mathbf{x} + \delta\mathbf{u}, \mathbf{y})] = \mathbb{E}_{\mathbf{u},\xi}[f(\mathbf{x} + \delta\mathbf{u}, \mathbf{y}; \xi)]. \tag{5}$$

Typical choices of $\mathbb{P}$ include the standard Gaussian distribution and the uniform distribution over the unit ℓ_2 ball. In this paper, we adopt the latter and derive the following results.

Lemma 4.2. *Let $f_\delta(\mathbf{x}, \mathbf{y}) := \mathbb{E}_{\mathbf{u}\sim\mathbb{P}}[f(\mathbf{x} + \delta\mathbf{u}, \mathbf{y})]$, where $\mathbb{P}$ is the uniform distribution over the unit ℓ_2 ball. Under Assumptions 3.1 and 5.1, the following properties hold:*

1. *$|f_\delta(\mathbf{x}, \mathbf{y}) - f(\mathbf{x}, \mathbf{y})| \leq \delta L$.*
2. *The function $f_\delta(\mathbf{x}, \mathbf{y})$ is differentiable and L-Lipschitz continuous, and its gradient $\nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y})$ is $\frac{cL\sqrt{d_1}}{\delta}$-Lipschitz continuous with respect to $\mathbf{x}$, where $c > 0$ is a constant.*
3. *For any $\mathbf{x} \in \mathbb{R}^{d_1}$, the function $f_\delta(\mathbf{x}, \cdot)$ is τ-strongly concave.*

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4. *The gradient $\nabla_{\mathbf{y}} f_\delta(\mathbf{x}, \mathbf{y})$ is L_{fy}-Lipschitz continuous with respect to $(\mathbf{x}, \mathbf{y})$.*

5. *Let $\partial_{\delta \mathbf{x}} f(\mathbf{x}, \mathbf{y})$ denote the Goldstein δ-subdifferential of f with respect to $\mathbf{x}$. Then, for any $\mathbf{x} \in \mathbb{R}^{d_1}$, we have $\nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}) \in \partial_{\delta \mathbf{x}} f(\mathbf{x}, \mathbf{y})$.*

For the smoothed minimax problem (5), we define its notion of ϵ-stationarity as follows.

Definition 4.3. A pair of points $(\mathbf{x}, \mathbf{y}) \in \mathbb{R}^{d_1} \times \mathcal{Y}$ is an ϵ-stationary point if it satisfies

$$\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y})\| \leq \epsilon, \ \|\mathcal{P}_{\mathcal{Y}}\left(\mathbf{y} + (1/L_{fy}) \nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y})\right) - \mathbf{y}\| \leq \epsilon / L_{fy}. \tag{6}$$

Then under Assumption 3.1 and 5.1, we can build the relationship between the stationarity notions for the original nonsmooth problem and its smoothed counterpart as follows.

Proposition 4.4. *Under Assumptions 3.1 and 5.1, if a pair $(\mathbf{x}, \mathbf{y})$ is an ϵ-stationary point of the smoothed objective $f_\delta(\mathbf{x}, \mathbf{y})$ in the sense of Definition 4.3, then $(\mathbf{x}, \mathbf{y})$ is a (δ, ϵ)-generalized Goldstein stationary point of the original objective $f(\mathbf{x}, \mathbf{y})$ as defined in Definition 4.1.*

Remark 4.5. Proposition 4.4 indicates that it suffices to design algorithms that find ϵ-stationary points of the smoothed minimax problem (5), which in turn guarantees convergence to a (δ, ϵ)-GGSP of the original nonsmooth minimax problem (1).

Algorithm 1 Zeroth- and First-Order Stochastic Gradient Descent-Ascent (ZOFOSGDA).

Initialize x_1, y_1.
Compute $\mathbf{v}_1 = \nabla_{\mathbf{y}} g(\mathbf{x}_1, \mathbf{y}_1; \mathcal{B}_1)$ and $\mathbf{w}_1 = \bar{\nabla}_{\mathbf{x}} f(\mathbf{x}_1, \mathbf{y}_1; \mathcal{B}_1)$
for $t = 1, \cdots, T$ **do**
  $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta_t \gamma \mathbf{w}_t$.
  $\tilde{\mathbf{y}}_{t+1} = \arg\min_{\mathbf{y} \in \mathcal{Y}} \{\langle \mathbf{v}_t, y \rangle + \frac{1}{2\lambda} \|\mathbf{y} - \mathbf{y}_t\|^2\}$ and $\mathbf{y}_{t+1} = \mathbf{y}_t + \eta_t(\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t)$.
  Compute $\mathbf{w}_{t+1} = (1 - \beta_{t+1}) \mathbf{w}_t + \beta_{t+1} \bar{\nabla}_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1}; \mathcal{B}_{t+1})$
  Compute $\mathbf{v}_{t+1} = (1 - \alpha_{t+1}) \mathbf{v}_t + \alpha_{t+1} \nabla_{\mathbf{y}} g(\mathbf{x}_{t+1}, \mathbf{y}_{t+1}; \mathcal{B}_{t+1})$
end for

Obviously, directly calculating the stochastic gradient estimition $\nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}; \mathcal{B})$ is impractical. However, an unbiased estimator of $\nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y})$ can be obtained using the function-query oracle calls, leading to a zeroth-order gradient estimator. Specifically, given a batch $\mathcal{B} = \{\xi_1, \ldots, \xi_M\}$ of *i.i.d.* samples from the underlying data distribution, we define

$$\bar{\nabla}_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}; \mathcal{B}) = \frac{1}{M} \sum_{i=1}^{M} \frac{d_1}{2\delta} (f(\mathbf{x} + \delta \mathbf{u}_i, \mathbf{y}; \xi_i) - f(\mathbf{x} - \delta \mathbf{u}_i, \mathbf{y}; \xi_i)) \mathbf{u}_i, \tag{7}$$

where each $\mathbf{u}_i$ is sampled independently from the uniform distribution over the unit ℓ_2 ball.

4.3 Proposed Method

In this subsection, we propose a hybrid zeroth- and first-order stochastic gradient descent–ascent method, termed ZOFOGDA. The detailed procedure is summarized in Algorithm 1. For notational convenience, we define $g(\mathbf{x}, \mathbf{y}) := -f(\mathbf{x}, \mathbf{y})$, so that $g(\mathbf{x}, \mathbf{y})$ is strongly convex with respect to $\mathbf{y}$.

In Algorithm 1, we incorporate an exponential moving average (EMA) scheme for both zeroth-order and first-order gradient estimates. The EMA mechanism helps smooth short-term stochastic fluctuations, capture long-term gradient trends, and reduce the variance inherent in stochastic gradient estimators [37, 32]. Specifically, the EMA updates are given by

$$\mathbf{w}_{t+1} = (1 - \beta_{t+1}) \mathbf{w}_t + \beta_{t+1} \bar{\nabla}_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1}; \mathcal{B}_{t+1}), \tag{8}$$

$$\mathbf{v}_{t+1} = (1 - \alpha_{t+1}) \mathbf{v}_t + \alpha_{t+1} \nabla_{\mathbf{y}} g(\mathbf{x}_{t+1}, \mathbf{y}_{t+1}; \mathcal{B}_{t+1}). \tag{9}$$

Here, $0 < \gamma < 1$ and $0 < \lambda < 1$ denote the step sizes for the gradient descent and ascent updates, respectively, and $0 < \eta_t < 1$ is an extrapolation parameter. Based on the EMA-smoothed gradient estimates, the updates for $\mathbf{x}$ and $\mathbf{y}$ are performed as

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta_t \gamma \mathbf{w}_t, \tag{10}$$

$$\tilde{\mathbf{y}}_{t+1} = \arg\min_{\mathbf{y} \in \mathcal{Y}} \left\{ \langle \mathbf{v}_t, \mathbf{y} \rangle + \frac{1}{2\lambda} \|\mathbf{y} - \mathbf{y}_t\|^2 \right\}, \tag{11}$$

$$\mathbf{y}_{t+1} = \mathbf{y}_t + \eta_t (\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t). \tag{12}$$

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By aggregating historical gradient information, the EMA scheme provides more stable momentum estimates for both variables, enabling consistent extrapolation steps and improving the robustness of the overall optimization procedure.

5 Convergence Analysis

In this section, we study the convergence properties of our proposed ZOFOGDA. All related proofs are provided in the following Appendix.

We first give the following additional assumptions, which are important for our method.

Assumption 5.1. [9] There exists a constant $L_{fxy} > 0$ such that for all $x \in \mathbb{R}^{d_1}$, and $\mathbf{y}, \mathbf{y}' \in \mathcal{Y}$, and all directions $\mathbf{d}$, the generalized Clarke directional derivative of f with respect to $\mathbf{x}$ satisfies $\left|f_x^\circ(\mathbf{x}, \mathbf{y}; \mathbf{d}) - f_x^\circ(\mathbf{x}, \mathbf{y}'; \mathbf{d})\right| \leq L_{fxy} \|\mathbf{y} - \mathbf{y}'\| \, \|\mathbf{d}\|$.

Assumption 5.2. The gradient estimation $\nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y}; \mathcal{B})$ are unbiased and have bounded variance, $\mathbb{E}[\nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y}; \mathcal{B})] = \nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y})$ and $\mathbb{E}[\|\nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y}; \mathcal{B}) - \nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y})\|^2] \leq \frac{\sigma_{\mathbf{y}}^2}{M}$.

Assumptions 5.2 are commonly used in optimization literatures [13, 40]. Assumption 5.1 introduced by [9] imposes a parameter-dependent stability condition on the generalized Clarke directional derivative of f with respect to $\mathbf{x}$. Although $f(\cdot, \mathbf{y})$ may be nonsmooth, this assumption ensures that the variation of the directional derivative with respect to changes in the parameter $\mathbf{y}$ is uniformly controlled, with a Lipschitz constant L_{fxy} is independent of the direction. Therefore, we have the following result.

Lemma 5.3. *Under Assumptions 3.1 and 5.1, we have $\nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y})$ is L_{fxy}-Lipschitz w.r.t $\mathbf{y}$.*

Lemma 5.4. *Under Assumption 3.1, we have $\mathbb{E}[\bar{\nabla}_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}; \mathcal{B})] = \nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y})$ and $\mathbb{E}[\|\bar{\nabla}_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}; \mathcal{B}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y})\|^2] \leq \sigma_{\mathbf{x}}^2$, where $\sigma_{\mathbf{x}}^2 = \frac{16\sqrt{2\pi} d_1 L^2}{M}$.*

According to [18], the smoothed minimax problem (5) is equivalent to the following minimization problem:

$$\min\{\Phi(\mathbf{x}) = \max_{\mathbf{y} \in \mathcal{Y}} f_\delta(\mathbf{x}, \mathbf{y}) = f_\delta(\mathbf{x}, \mathbf{y}_\delta^*(\mathbf{x}))\}, \tag{13}$$

where $y_\delta^*(\mathbf{x}) = \arg\max_{\mathbf{y} \in \mathcal{Y}} f_\delta(\mathbf{x}, \mathbf{y})$. Under the given Assumptions 3.1 and 5.1, we can obtain the following results.

Lemma 5.5. *Under Assumptions 3.1 and 5.1, we can obtain $\Phi(\mathbf{x})$ is ℓ-smooth and $\nabla\Phi(\mathbf{x}) = \nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}_\delta^*(\mathbf{x}))$, where $\ell = \frac{cL\sqrt{d_1}}{\delta} + L_{fxy}\kappa$ and $\kappa = \frac{L_{fy}}{\tau}$.*

We can also define the ϵ-stationary of problem (13) as follows:

Definition 5.6. A point x is an ϵ-stationary point of function Φ if $\|\nabla\Phi(\mathbf{x})\| \leq \epsilon$.

Then, we can build the relationship between Definition 4.3 and Definiton 5.6 as follows.

Proposition 5.7. *Under Assumptions 3.1 and 5.1, if a point $\hat{\mathbf{x}}$ is an ϵ-stationary point in terms of Definition 5.6, an $\mathcal{O}(\epsilon)$-stationary point $(\mathbf{x}', \mathbf{y}')$ in terms of Definition 4.3 can be obtained using $\delta = \mathcal{O}(\epsilon)$ and additional $\mathcal{O}(\epsilon^{-2})$ stochastic gradients. Conversely, if a point $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ is an ϵ-stationary point in terms of Definition 4.3, a point $\hat{\mathbf{x}}$ is an $\mathcal{O}(\epsilon)$-stationary point in terms of Definition 5.6 with $\delta = \mathcal{O}(\epsilon)$.*

Remark 5.8. Based on Proposition 4.4 and 5.7 and Lemma 4.2, we can conclude that if we can find the ϵ-statioanry of $\Phi(\mathbf{x})$ defined in Definition 5.6, then we can obtain the (δ, ϵ)-GGSP of the nonsmooth minimax problem defined in Definiton 4.1.

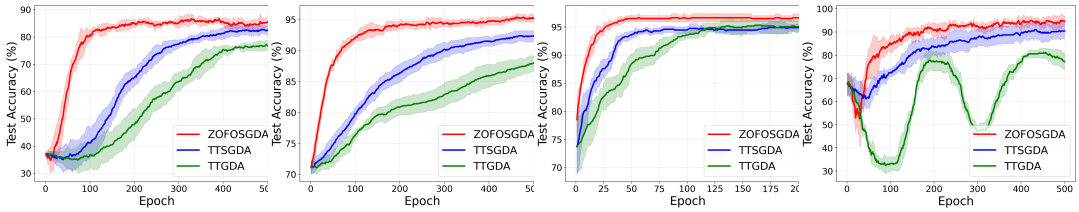
In addition, we assume the function (13) is lower bounded as follows,

Assumption 5.9. The function $\Phi(\mathbf{x})$ is bounded below in $\mathbf{R}^{d_1}$, *i.e.,* $\Phi^* = \inf_x \Phi(\mathbf{x}) > -\infty$.

Under all the above assumptions, the convergence properties of ZOFOGDA are established in the following result (A more detailed version is presented in our Appendix).

Theorem 5.10. *Under Assumptions 3.1, 5.1, 5.2, and 5.9, in Algorithm 1, let $\eta_t = \frac{k}{(m+t)^{1/2}}$ for all $t \geq 1$, $\ell = \mathcal{O}(d_1^{1/2}\delta^{-1})$, $k = \mathcal{O}(1)$,$m = \mathcal{O}(\delta^{-1})$, $\lambda^{-1} = \mathcal{O}(\tau)$, $\gamma^{-1} = \mathcal{O}((d_1/\delta)^{1/2})$, $\delta = \mathcal{O}(\epsilon)$,*

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(a) Australian (b) Svmguide1 (c) Breast-cancer (d) Ionosphere_scale

Figure 1: Test accuracy against training time of all the methods in robust classification.

$T = \tilde{O}(d_1\delta^{-1}\epsilon^{-4})$ and $M = \mathcal{O}(\sqrt{d_1})$, our method outputs a (δ, ϵ)-GGSP and the zeroth-order oracle complexity is

$$\tilde{\mathcal{O}}(d_1^{1.5}\delta^{-1}\epsilon^{-4}\}). \tag{14}$$

Table 2: Test accuracy (%) with standard variance of all the methods in robust classification.

Methods	Australian	Svmguide1	Breast-cancer	Ionosphere_scale
TTGDA	76.81 ± 1.89	87.96 ± 1.32	95.04 ± 1.07	77.18 ± 3.72
TTSGDA	82.46 ± 1.68	92.36 ± 0.85	94.89 ± 0.65	90.42 ± 3.26
ZOFOSGDA	**85.51** ± 1.21	**95.28** ± 0.66	**96.50** ± 0.55	**94.65** ± 2.42

6 Experiments

In this section, we evaluate the performance of our method on two representative tasks: on AUC maximization and robust classification. All experiments are conducted on a workstation equipped with four NVIDIA 4090 Ti GPUs.

6.1 Robust Classification

In this subsection, we consider robust classification with nonconvex penalty functions [19]. Given a dataset $\{(\boldsymbol{a}_i, b_i)\}_{i=1}^N$, where $\boldsymbol{a}_i \in \mathbb{R}^{d_1}$ denotes the feature vector of the i-th sample and $b_i \in \{+1, -1\}$ is the corresponding label, the problem can be formulated as the following minimax optimization:

$$f(\mathbf{x}, \mathbf{y}) = \frac{1}{N}\sum_{i=1}^{N} y_i \ell(M_{\mathbf{x}}(\boldsymbol{a}_i), b_i) - \frac{\theta_1}{2}\left\|N\mathbf{y} - \mathbf{1}_{d_2}\right\|^2 \tag{15}$$

where $M_{\mathbf{x}}(\cdot)$ denotes the model parameterized by $\mathbf{x}$, $\ell(\cdot, \cdot)$ denotes the loss function, and the variable $\mathbf{y}$ represents sample weights constrained to the probability simplex $\mathcal{Y} = \{\mathbf{y} \in \mathbb{R}_+^{d_2} : \mathbf{y}^\top \mathbf{1}_{d_2} = 1\}$. The quadratic regularization term on $\mathbf{y}$ encourages the weights to remain close to the uniform distribution.

In our experiments, we adopt a simple one-hidden-layer MLP with ReLU activations and use hinge loss for training. We evaluate our proposed ZOFOSGDA by comparing it with TTSGDA [19] and TTGDA [19] on several benchmark datasets from LIBSVM [1]. Unless otherwise stated, we fix $\theta_1 = 0.01$ throughout. Following the experimental protocol in [19], for both TTSGDA and TTGDA, we tune the step size of the $\mathbf{y}$-update by selecting $\eta_y \in \{1, 0.1, 0.01\}$, and we further vary the relative scaling between the $\mathbf{x}$ and $\mathbf{y}$ updates by setting $\eta_y/\eta_x \in \{10, 100, 1000\}$. For our method, we use a time-varying stepsize schedule $\eta_t = \frac{10}{\sqrt{1000+t}}$ and set $c_1 = c_2 = 3$. In addition, we tune the hyperparameters by choosing $\lambda \in \{1, 0.1, 0.01\}$ and $\gamma \in \{1, 0.1, 0.01\}$. We also choose $M = 5$ to calculate the zeroth-order in our method. For all the methods, we set the batch size at 128.All methods are run under the same model architecture and data preprocessing to ensure a fair comparison with three different seeds.

[1] `https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets/`

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Table 3: Comparison of the testing AUC values (mean±std).

Algorithm	Climate	German	Svmguide1	Spambase	A9a	Ijcnn1
TTGDA	0.878±0.014	**0.778**±0.006	0.694±0.004	0.956±0.003	0.895±0.001	0.881±0.002
TTSGDA	0.881±0.007	0.754±0.014	0.698±0.022	**0.968**±0.001	0.876±0.002	0.882±0.006
ZOFOSGDA	**0.888**±0.003	0.774±0.011	**0.708**±0.006	**0.968**±0.005	**0.899**±0.001	**0.884**±0.005

The results are reported in Table 2 and Fig. 3. Across most LIBSVM datasets, ZOFOSGDA attains the best (or highly competitive) test accuracy, and it reaches its final performance in substantially fewer iterations than TTSGDA and TTGDA. We attribute this gap primarily to the two-timescale design of TTSGDA and TTGDA. To ensure stable primal–dual interaction, these methods typically require a strict stepsize separation, i.e., $\eta_x \ll \eta_y$. While such a choice helps control the dynamics, it also makes the $\mathbf{x}$-updates overly conservative, which can significantly slow down the overall convergence and delay improvements in test accuracy. In contrast, ZOFOSGDA does not rely on a stringent timescale separation and thus can make more effective progress per iteration. Moreover, our momentum mechanism smooths the update directions and reduces oscillations induced by stochasticity, yielding more stable training and better solution quality. Together, these factors explain why ZOFOSGDA consistently achieves both faster convergence and higher test accuracy than TTSGDA and TTGDA in this experiment.

6.2 AUC Maximization

In this subsection, we perform experiments on the problem of AUC maximization [46] with the least square loss to evaluate the performance of our ZOFOSGDA. In this case, AUC maximization can be formulated as

$$\min_{\mathbf{x}=(\theta,\mu_1,\mu_2)} \max_{\mathbf{y}} \mathbb{E}_{(\boldsymbol{a},b)}[f(\theta,\mu_1,\mu_2,\mathbf{y};\boldsymbol{a},b)] \tag{16}$$

where $f = (1-p)\big(h(\theta;\boldsymbol{a})-\mu_1\big)^2\mathbb{I}[b=1] + p\big(h(\theta;\boldsymbol{a})-\mu_2\big)^2\mathbb{I}[b=-1] + 2(1+\mathbf{y})\Big(p\,h(\theta;\boldsymbol{a})\mathbb{I}[b=-1] - (1-p)h(\theta;\boldsymbol{a})\mathbb{I}[b=1]\Big) - p(1-p)\mathbf{y}^2$ and $p = \mathbb{P}(b=1)$, $(\boldsymbol{a},b)$ is the data sample. In this experiment, we use a one-hidden-layer MLP with ReLU activations as the model. We evaluate ZOFOSGDA by comparing it with TTSGDA [19] and TTGDA [19] on the datasets listed in Table 5. All methods were tested on models with three different initializations. For other settings, we follow the setting presented in the above subsection.

Table 3 reports the testing AUC (mean ± std) of different methods on several benchmark datasets. As shown in the table, the proposed method (ZOFOSGDA) achieves the best or highly competitive AUC performance across all reported datasets. In particular, ZOFOSGDA attains the highest AUC on Climate, Svmguide1, A9a, and Ijcnn1, and matches the best performance on Spambase. Compared with TTGDA and TTSGDA, the proposed method consistently yields higher mean AUC values while maintaining comparable or smaller standard deviations on most datasets, indicating more stable performance. Overall, these results demonstrate that ZOFOSGDA provides consistent improvements in classification accuracy while preserving robustness across different datasets.

7 Conclusion

In this work, we study nonsmooth minimax optimization problems and propose a hybrid zeroth- and first-order descent–ascent algorithm equipped with momentum updates. Leveraging the generalized Goldstein subdifferential, we establish non-asymptotic convergence guarantees and show that the proposed method achieves a query complexity of $\tilde{\mathcal{O}}(d_1^{1.5}\delta^{-1}\epsilon^{-4})$, matching the best-known rates for zeroth-order nonsmooth minimization while extending them to the more challenging minimax setting. From a theoretical perspective, our analysis provides new insights into the role of zeroth-order information and momentum in handling nonsmooth and nonconvex–nonconcave structures. Empirically, experiments on two applications demonstrate that our method outperforms competitive first-order baselines, highlighting its robustness and practical efficiency.

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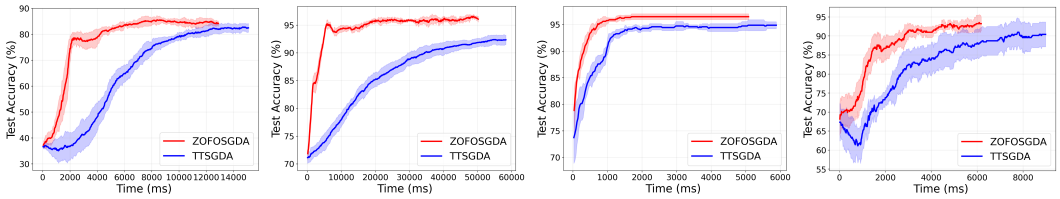
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A Additional Experiments

A.1 Training time in Robust Classification.

We compare the training time of our method with TTSGDA and present the result in Fig 3.



(a) Australian (b) Svmguide1 (c) Breast-cancer (d) Ionosphere scale

Figure 3: Test accuracy against training time of all the methods in robust classification.

A.2 Ablation Study in Robust Classification.

As shown in Table 4, the classification accuracy is influenced by both the coefficient value (c_1 or c_2) and the batch size in robust classification. Overall, smaller coefficient values lead to more stable performance across different datasets, while larger values tend to introduce higher sensitivity to the batch size. When c_1 (or equivalently c_2) is set to 1 or 2, the proposed method achieves consistently strong results on all datasets. In particular, a batch size of 128 yields the best or near-best performance in most cases, indicating a favorable trade-off between optimization stability and generalization. These results demonstrate that the ZOFOSGDA is robust to the choice of batch size when moderate coefficient values are used, and that setting c_1 and c_2 to relatively small values provides a more reliable performance across datasets.

Table 4: Performance of our method with different c_1, c_2 and batch size in robust classification.

c_1&c_2	batch_size	Australian	Svmguide1	Breast-cancer	Ionosphere_scale
1	64	84.06 ± 1.89	95.92 ± 0.64	96.64 ± 0.74	90.70 ± 2.90
1	128	84.20 ± 2.57	95.79 ± 0.46	96.50 ± 0.55	92.68 ± 1.64
1	256	81.74 ± 1.25	95.73 ± 0.46	96.50 ± 0.55	90.14 ± 3.67
2	64	85.80 ± 1.49	95.31 ± 0.61	96.64 ± 0.74	92.68 ± 3.38
2	128	84.49 ± 2.27	95.44 ± 0.60	96.50 ± 0.55	94.65 ± 2.42
2	256	84.35 ± 1.08	95.24 ± 0.53	96.50 ± 0.55	91.55 ± 3.09
3	64	85.22 ± 1.49	95.11 ± 0.39	96.64 ± 0.74	93.52 ± 3.03
3	128	85.51 ± 1.21	95.28 ± 0.66	96.50 ± 0.55	94.65 ± 2.42
3	256	84.78 ± 1.45	95.18 ± 0.70	96.50 ± 0.55	92.11 ± 3.52

A.3 Datasets in AUC Maximization

We evaluate our ZOFOGDA on nine benchmark binary classification datasets with varying scales, feature dimensions, and class imbalance levels. The statistics of these datasets are summarized in Table 5.

Specifically, the datasets range from small-scale problems such as Climate (540 instances) to large-scale datasets such as Acoustic with over 78k instances. The feature dimensionality varies from 4 (svmguide1) to 126 (Connect), covering both low- and high-dimensional settings. In addition, the imbalance rates span a wide range, from relatively balanced datasets (e.g., Climate and Ijcnn1) to highly imbalanced ones such as Covtype, making them suitable for evaluating robustness under class imbalance.

Overall, this diverse collection of datasets allows us to comprehensively assess the effectiveness and scalability of the proposed method across different data characteristics.

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Table 5: Statistics of Experimental Datasets

ID	Datasets	Instances	Features	Imbalance Rate
1	climate	540	18	0.085
2	german	1,000	24	0.302
3	svmguide1	3,089	4	0.353
4	spambase	4,601	57	0.394
5	a9a	32,561	123	0.240
6	ijcnn1	49,990	22	0.097

A.4 Ablation Study in AUC Maximization

Table 6 reports the ablation results with respect to $c_1\&c_2$ and batch size. Overall, the performance remains stable across different parameter settings, indicating that the proposed method is not sensitive to these hyperparameters in AUC maximization. On the Climate and German datasets, the accuracy varies only slightly under different configurations. For Svmguide1, a larger batch size (256) consistently yields better performance, while the results on Spambase are nearly identical across all settings with very small standard deviations, demonstrating the robustness and stability of the proposed method.

Table 6: Ablation study on accuracy

$c_1\&c_2$	batch_size	climate	german	svmguide1	spambase
1	64	0.874 ± 0.016	0.781 ± 0.01	0.688 ± 0.008	0.966 ± 0.003
1	128	0.876 ± 0.019	0.767 ± 0.013	0.690 ± 0.006	0.961 ± 0.003
1	256	0.873 ± 0.014	0.767 ± 0.008	0.706 ± 0.007	0.958 ± 0.004
2	64	0.875 ± 0.016	0.781 ± 0.009	0.688 ± 0.008	0.966 ± 0.003
2	128	0.876 ± 0.019	0.766 ± 0.013	0.69 ± 0.006	0.961 ± 0.003
2	256	0.874 ± 0.014	0.766 ± 0.008	0.706 ± 0.008	0.958 ± 0.004
3	64	0.874 ± 0.017	0.78 ± 0.008	0.688 ± 0.008	0.966 ± 0.003
3	128	0.877 ± 0.021	0.764 ± 0.011	0.69 ± 0.005	0.961 ± 0.003
3	256	0.873 ± 0.013	0.765 ± 0.008	0.706 ± 0.008	0.958 ± 0.004

A.5 Meta-Training in LLMs

In this subsection, we consider a large-scale meta-learning task inspired by Objective (5) in [33], which arises naturally in meta-learning for large language models. In this setting, the goal is to learn shared model parameters that generalize well across tasks, while accounting for the variability induced by task-specific fine-tuning.

Following [33], the meta-learning problem is formulated as a minimax optimization that explicitly models different task-level adaptations through auxiliary fine-tuning parameters. Let $\mathcal{D}_f$ denote a set of evaluation (query) tasks and $\mathcal{D}_p$ denote a set of support tasks. We consider the following objective:

$$\min_{\mathbf{x},\hat{\mathbf{y}}} \max_{\mathbf{y}} G_\theta(\mathbf{x},\hat{\mathbf{y}},\mathbf{y}) := F(\mathbf{x},\hat{\mathbf{y}};\mathcal{D}_f) + \theta\big(F(\mathbf{x},\hat{\mathbf{y}};\mathcal{D}_p) - F(\mathbf{x},\mathbf{y};\mathcal{D}_p)\big), \tag{17}$$

where $F(\mathbf{x},\mathbf{y};\mathcal{D})$ denotes the average task loss of model parameters $\mathbf{x}$ after fine-tuning with task-specific parameters $\mathbf{y}$ on dataset $\mathcal{D}$, and $\theta > 0$ balances the meta-level generalization objective and the task-level fitting behavior.

In this formulation, both $\hat{\mathbf{y}}$ and $\mathbf{y}$ represent fine-tuning parameters applied to the shared model $\mathbf{x}$, but they play different roles. The parameter $\hat{\mathbf{y}}$ corresponds to a controlled fine-tuning strategy that is optimized jointly with $\mathbf{x}$, whereas $\mathbf{y}$ represents an alternative task adaptation that may lead to suboptimal generalization on the evaluation tasks. The inner maximization over $\mathbf{y}$ thus characterizes the most challenging task-level fine-tuning behavior under the current model, while the outer minimization seeks shared parameters $\mathbf{x}$ and fine-tuning $\hat{\mathbf{y}}$ that remain robust to such variations.

In this experiment, following the experimental protocol of [33], we evaluate our approach on a range of classification and multiple-choice tasks. In this setting, training and testing are conducted on the

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same task. We employ LoRA [10], which are widely used parameter-efficient fine-tuning (PEFT) baselines, to parameterize both $\hat{\mathbf{y}}$ and $\mathbf{y}$. We adopt OPT-1.3B as the pretrained backbone model. For each task, we use 500 examples for validation and 1,000 examples for testing. We compare our method against several baselines, including MeZO [25] for full-model fine-tuning, MeZO for PEFT, and first-order PEFT methods. We report test accuracy or F1 score as the evaluation metric. We train our methods for 2,000 steps. Due to resource constraints, we sweep only γ and λ as hyperparameters. Specifically, $\gamma \in \{10^{-2}, 10^{-3}, 10^{-4}\}$ and $\lambda \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$. We set $k = 1.0$, $m = 1000.0$, $c_1 = 31.0$, and $c_2 = 31.0$ such that $1 - \alpha$ and $1 - \beta$ gradually increase from 0.01. Unless otherwise specified, we follow the same experimental settings as in [33]. We perform all experiments in table 7 using three random seeds and report the average. We can find that our method can achieve similar results as Bilevl-ZOFO [33]. These results also demonstrate the superiority of our method.

Table 7: Single-Task Experiments on OPT-1.3B with 1000 samples. Values correspond to mean across three random seeds. FO: First-Order. FT: full-model fine-tuning.

Trainer	Mode	BoolQ	CB	Copa	RTE	SST2	WIC	WinoGrande	WSC	Average
FO	lora	0.7456	0.8512	0.7500	0.7292	0.9258	0.6463	0.5806	0.6474	0.7345
MeZO	ft	0.6927	0.7767	0.7000	0.6587	0.8214	0.5543	0.5480	0.5054	0.6572
	lora	0.6860	0.7607	0.7200	0.6755	0.8501	0.5549	0.5607	0.5570	0.6706
Bilevel-ZOFO	lora	**0.7433**	**0.9167**	0.7400	0.7401	**0.9331**	0.6447	0.5903	**0.6428**	0.7439
Ours	lora	0.7407	0.8750	**0.8067**	**0.7593**	0.9312	**0.6510**	**0.6007**	0.6314	**0.7495**

B Details of Proofs

B.1 Proofs of Lemma 4.2

Proof. Let $\mathbf{u} \in \mathbb{R}^{d_1}$ denote a random variable distruted unifromly on $\mathbb{B}_1(0)$. For the first statement, since $f(\mathbf{x}, \mathbf{y})$ is L-Lipschitz, we have

$$|f_\delta(\mathbf{x}, \mathbf{y}) - f(\mathbf{x}, \mathbf{y})| = |\mathbb{E}_{\mathbf{u}\sim\mathbb{P}}[f(\mathbf{x} + \delta\mathbf{u}, \mathbf{y})] - f(\mathbf{x}, \mathbf{y})| \leq \delta L \mathbb{E}[\|\mathbf{u}\|] \leq \delta L \tag{18}$$

Then, we proceed to prove the second statement. Since f is L-Lipschitz, we have

$$\begin{aligned}
&|f_\delta(\mathbf{x}, \mathbf{y}) - f_\delta(\mathbf{x}', \mathbf{y}')| \\
=&|f_\delta(\mathbf{x}, \mathbf{y}) - f_\delta(\mathbf{x}', \mathbf{y})| + |f_\delta(\mathbf{x}', \mathbf{y}) - f_\delta(\mathbf{x}', \mathbf{y}')| \\
=&|\mathbb{E}[f(\mathbf{x} + \delta\mathbf{u}, \mathbf{y}) - f(\mathbf{x}' + \delta\mathbf{u}, \mathbf{y})]| + |\mathbb{E}[f(\mathbf{x} + \delta\mathbf{u}, \mathbf{y}) - f(\mathbf{x} + \delta\mathbf{u}, \mathbf{y}')]| \\
\leq& L(\|\mathbf{x} - \mathbf{x}'\| + \|\mathbf{y} - \mathbf{y}'\|)
\end{aligned} \tag{19}$$

It remains to prove that $\nabla_{\mathbf{x}} f(\mathbf{x}, \mathbf{y})$ is Lipschitz. Since f is L-Lipschitz, Rademacher's theorem guarantees that f is almost everywhere differentiable. This implies that $\nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}) = \mathbb{E}[\nabla_{\mathbf{x}} f(\mathbf{x} + \delta\mathbf{u}, \mathbf{y})]$ and $\nabla_{\mathbf{y}} f_\delta(\mathbf{x}, \mathbf{y}) = \mathbb{E}[\nabla_{\mathbf{y}} f(\mathbf{x} + \delta\mathbf{u}, \mathbf{y})]$. Then, we have

$$\begin{aligned}
&\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}', \mathbf{y})\| \\
=&\|\mathbb{E}[\nabla_{\mathbf{x}} f(\mathbf{x} + \delta\mathbf{u}, \mathbf{y})] - \mathbb{E}[\nabla_{\mathbf{x}} f(\mathbf{x}' + \delta\mathbf{u}, \mathbf{y})]\| \\
=&\frac{1}{\mathrm{Vol}(\mathbb{B}_1(0))} \left| \int_{u\in\mathbb{B}_1(0)} \nabla_{\mathbf{x}} f(\mathbf{x} + \delta\mathbf{u}, \mathbf{y}) d\mathbf{u} - \int_{u\in\mathbb{B}_1(0)} \nabla_{\mathbf{x}} f(\mathbf{x}' + \delta\mathbf{u}, \mathbf{y}) d\mathbf{u} \right| \\
=&\frac{1}{\mathrm{Vol}(\mathbb{B}_\delta(0))} \left| \int_{z\in\mathbb{B}_\delta(\mathbf{x})} \nabla_{\mathbf{x}} f(\mathbf{z}, \mathbf{y}) d\mathbf{z} - \int_{z\in\mathbb{B}_\delta(\mathbf{x}')} \nabla_{\mathbf{x}} f(\mathbf{z}, \mathbf{y}) d\mathbf{z} \right|
\end{aligned} \tag{20}$$

Since f is L-Lipschitz, we have $\|\nabla_{\mathbf{x}} f(\mathbf{x}, \mathbf{y})\| \leq L$ for any $x \in \mathbb{B}_\delta(\mathbf{x}) \bigcup \mathbb{B}_\delta(\mathbf{x}')$. Then, if $\|\mathbf{x} - \mathbf{x}'\| \geq 2\delta$, we have

$$\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}', \mathbf{y})\| \leq 2L \leq \frac{L\|\mathbf{x} - \mathbf{x}'\|}{\delta} \leq \frac{L\sqrt{d_1}\|\mathbf{x} - \mathbf{x}'\|}{\delta}. \tag{21}$$

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If $\|\mathbf{x}-\mathbf{x}'\| \leq 2\delta$, it is clear that $\mathbb{B}_\delta(\mathbf{x}) \bigcap \mathbb{B}_\delta(\mathbf{x}') \neq \emptyset$. Then, we can obtain

$$\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x},\mathbf{y}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}',\mathbf{y})\| \leq \frac{1}{\mathrm{Vol}(\mathbb{B}_\delta(0))} \left| \int_{z\in\mathbb{B}_\delta(\mathbf{x})\backslash\mathbb{B}_\delta(\mathbf{x}')} \nabla_{\mathbf{x}} f(\mathbf{z},\mathbf{y}) d\mathbf{z} - \int_{z\in\mathbb{B}_\delta(\mathbf{x}')\backslash\mathbb{B}_\delta(\mathbf{x})} \nabla_{\mathbf{x}} f(\mathbf{z},\mathbf{y}) d\mathbf{z} \right| \tag{22}$$

Since $\|\nabla_{\mathbf{x}} f(\mathbf{x},\mathbf{y})\| \leq L$, we have

$$\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x},\mathbf{y}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}',\mathbf{y})\| \leq \frac{L}{\mathrm{Vol}(\mathbb{B}_\delta(0))} \left(\mathrm{Vol}(\mathbb{B}_\delta(\mathbf{x})\backslash\mathbb{B}_\delta(\mathbf{x}')) - \mathrm{Vol}(\mathbb{B}_\delta(\mathbf{x}')\backslash\mathbb{B}_\delta(\mathbf{x}))\right). \tag{23}$$

By the symmetry from a geometrical point of view, we have $\mathrm{Vol}(\mathbb{B}_\delta(\mathbf{x})\backslash\mathbb{B}_\delta(\mathbf{x}')) = \mathrm{Vol}(\mathbb{B}_\delta(\mathbf{x}')\backslash\mathbb{B}_\delta(\mathbf{x}))$. For simplicity, let $I = \mathbb{B}_\delta(\mathbf{x})\backslash\mathbb{B}_\delta(\mathbf{x}')$ and we have

$$\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x},\mathbf{y}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}',\mathbf{y})\| \leq \frac{2L}{\mathrm{Vol}(\mathbb{B}_\delta(0))}\mathrm{Vol}(I) \leq \frac{cL\sqrt{d_1}\|\mathbf{x}-\mathbf{x}'\|}{\delta} \tag{24}$$

where $c > 0$ is a constant. Therefore, we can conclude that

$$\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x},\mathbf{y}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}',\mathbf{y})\| \leq \frac{cL\sqrt{d_1}}{\delta}\|\mathbf{x}-\mathbf{x}'\| \tag{25}$$

For the third statement, since $f(\mathbf{x},\cdot)$ is τ-strongly concave, we have

$$f(\mathbf{x},\lambda\mathbf{y}_1 + (1-\lambda)\mathbf{y}_2) \geq \lambda f(\mathbf{x},\mathbf{y}_1) + (1-\lambda) f(\mathbf{x},\mathbf{y}_2) - \frac{\tau}{2}\lambda(1-\lambda)\|\mathbf{y}_1-\mathbf{y}_2\|^2 \tag{26}$$

Then, we have

$$\begin{aligned} f_\delta(\mathbf{x},\lambda\mathbf{y}_1 + (1-\lambda)\mathbf{y}_2) =& \mathbb{E}[f(\mathbf{x}+\delta\mathbf{u},\lambda\mathbf{y}_1 + (1-\lambda)\mathbf{y}_2)] \\ \geq& \mathbb{E}[\lambda f(\mathbf{x}+\delta\mathbf{u},\mathbf{y}_1) + (1-\lambda) f(\mathbf{x}+\delta\mathbf{u},\mathbf{y}_2) - \frac{\tau}{2}\lambda(1-\lambda)\|\mathbf{y}_1-\mathbf{y}_2\|^2] \\ =& \lambda f_\delta(\mathbf{x},\mathbf{y}_1) + (1-\lambda) f_\delta(\mathbf{x},\mathbf{y}_2) - \frac{\tau}{2}\lambda(1-\lambda)\|\mathbf{y}_1-\mathbf{y}_2\|^2 \end{aligned} \tag{27}$$

For the fourth statement, we have

$$\begin{aligned} &\|\nabla_{\mathbf{y}} f_\delta(\mathbf{x}_1,\mathbf{y}_1) - \nabla_{\mathbf{y}} f_\delta(\mathbf{x}_2,\mathbf{y}_2)\| \\ \leq& \|\nabla_{\mathbf{y}} f_\delta(\mathbf{x}_1,\mathbf{y}_1) - \nabla_{\mathbf{y}} f_\delta(\mathbf{x}_1,\mathbf{y}_2)\| + \|\nabla_{\mathbf{y}} f_\delta(\mathbf{x}_1,\mathbf{y}_2) - \nabla_{\mathbf{y}} f_\delta(\mathbf{x}_2,\mathbf{y}_2)\| \\ \leq& \|\mathbb{E}_{\mathbf{u}}[\nabla_{\mathbf{y}} f(\mathbf{x}_1+\delta\mathbf{u},\mathbf{y}_1) - \nabla_{\mathbf{y}} f(\mathbf{x}_1+\delta\mathbf{u},y_2)]\| + \|\mathbb{E}_{\mathbf{u}}[\nabla_{\mathbf{y}} f(\mathbf{x}_1+\delta\mathbf{u},\mathbf{y}_2) - \nabla_{\mathbf{y}} f(\mathbf{x}_2+\delta\mathbf{u},\mathbf{y}_2)]\| \\ \leq& \mathbb{E}_{\mathbf{u}}[\|\nabla_{\mathbf{y}} f(x_1+\delta\mathbf{u},y_1) - \nabla_{\mathbf{y}} f(\mathbf{x}_1+\delta\mathbf{u},\mathbf{y}_2)\|] + \mathbb{E}_{\mathbf{u}}[\|\nabla_{\mathbf{y}} f(\mathbf{x}_1+\delta\mathbf{u},\mathbf{y}_2) - \nabla_{\mathbf{y}} f(x_2+\delta\mathbf{u},y_2)\|] \\ \leq& \mathbb{E}_{\mathbf{u}}[L_{fy}\|\mathbf{y}_1-\mathbf{y}_2\|] + \mathbb{E}_{\mathbf{u}}[L_{fy}\|\mathbf{x}_1-\mathbf{x}_2\|] \\ =& L_{fy}(\|\mathbf{y}_1-\mathbf{y}_2\| + \|\mathbf{x}_1-\mathbf{x}_2\|). \end{aligned} \tag{28}$$

This completes the proof.

□

B.2 Proof of Proposition 4.4

Proof. Assume that a point $(\mathbf{x}',\mathbf{y}')$ satisfies the following condition,

$$\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x}',\mathbf{y}')\| \leq \epsilon \tag{29}$$

$$\|\mathcal{P}_{\mathcal{Y}}\left(\mathbf{y}' + (1/L_{fy})\nabla_{\mathbf{y}} f(\mathbf{x}',\mathbf{y}')\right) - \mathbf{y}'\| \leq \epsilon/L_{fy} \tag{30}$$

Since $\nabla_{\mathbf{x}} f_\delta(\mathbf{x}',\mathbf{y}') \in \partial_{\delta\mathbf{x}} f(\mathbf{x}',\mathbf{y}')$, we have

$$\min\{\|\mathbf{h}\| : \mathbf{h} \in \partial_{\delta\mathbf{x}} f(\mathbf{x}',\mathbf{y}')\} \leq \epsilon. \tag{31}$$

That completes the proof. □

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B.3 Proof of Lemma 5.3

Proof. Since $f(\cdot, \mathbf{y})$ is Lipschitz *w.r.t.* $\mathbf{x}$, its generalized Clarke directional derivative $f^\circ_{\mathbf{x}}(\mathbf{x}, \mathbf{y}; \mathbf{d})$ exists for all $\mathbf{x}, \mathbf{y}$ and all directions $\mathbf{d}$. It is a standard result of random smoothing that the function $f_\delta(\cdot, \mathbf{y})$ is continuously differentiable in $\mathbf{x}$, even when $f(\cdot, \mathbf{y})$ is nonsmooth, and that its directional derivative satisfies

$$\langle \nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}), \mathbf{d} \rangle = \mathbb{E}_{\mathbf{u}}[f^\circ_{\mathbf{x}}(\mathbf{x} + \delta\mathbf{u}, \mathbf{y}; \mathbf{d})], \qquad \forall \mathbf{d}.$$

Let y, y' be arbitrary. For any direction $\mathbf{d}$, we have

$$\begin{aligned}
\left|\langle \nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}'), \mathbf{d} \rangle\right| &= \left|\mathbb{E}_{\mathbf{u}}[f^\circ_{\mathbf{x}}(\mathbf{x} + \delta\mathbf{u}, \mathbf{y}; \mathbf{d}) - f^\circ_{\mathbf{x}}(\mathbf{x} + \delta\mathbf{u}, \mathbf{y}'; \mathbf{d})]\right| \\
&\leq \mathbb{E}_{\mathbf{u}}\left[\left|f^\circ_{\mathbf{x}}(\mathbf{x} + \delta\mathbf{u}, \mathbf{y}; \mathbf{d}) - f^\circ_{\mathbf{x}}(\mathbf{x} + \delta\mathbf{u}, \mathbf{y}'; \mathbf{d})\right|\right] \\
&\leq L_{fxy} \left\|\mathbf{y} - \mathbf{y}'\right\| \left\|\mathbf{d}\right\|,
\end{aligned}$$

where the last inequality follows from the assumption.

Since the above inequality holds for all directions $\mathbf{d}$, taking the supremum over all $\|\mathbf{d}\| = 1$ and using the dual norm representation yields

$$\left\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}')\right\| = \sup_{\|\mathbf{d}\|=1} \left|\langle \nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}'), d \rangle\right| \leq L_{fxy} \left\|\mathbf{y} - \mathbf{y}'\right\|.$$

This completes the proof. □

B.4 Proof of Lemma 5.4

Proof. By the definition of $\bar{\nabla}_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}; \mathcal{B})$ and the symmetry of the distribution of $\mathbf{u}$, we have

$$\begin{aligned}
&\mathbb{E}[\bar{\nabla}_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}; \mathcal{B})] \\
=&\mathbb{E}[\frac{1}{M}\sum_{i=1}^{M} \frac{d_1}{2\delta}(f(\mathbf{x} + \delta\mathbf{u}_i, \mathbf{y}; \xi_i) - f(\mathbf{x} - \delta\mathbf{u}_i, \mathbf{y}; \xi_i))\mathbf{u}_i] \\
=&\frac{1}{2M}\sum_{i=1}^{M}\left(\mathbb{E}[\frac{d_1}{\delta} f(\mathbf{x} + \delta\mathbf{u}_i, \mathbf{y}; \xi_i)\mathbf{u}_i] + \mathbb{E}[\frac{d_1}{\delta} f(\mathbf{x} - \delta\mathbf{u}_i, \mathbf{y}; \xi_i)(-\mathbf{u}_i)]\right) \\
=&\frac{1}{2M}\sum_{i=1}^{M}(\nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}) + \nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y})) \\
=&\nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y})
\end{aligned} \tag{32}$$

Since $\bar{\nabla}_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}; \mathcal{B}) = \frac{1}{M}\sum_{i=1}^{M} \frac{d_1}{2\delta}(f(\mathbf{x} + \delta\mathbf{u}_i, \mathbf{y}; \xi_i) - f(\mathbf{x} - \delta\mathbf{u}_i, \mathbf{y}; \xi_i))\mathbf{u}_i = \frac{1}{M}\sum_{i=1}^{M} g_i$, We have

$$\begin{aligned}
&\mathbb{E}\left[\|\bar{\nabla}_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y}; \mathcal{B}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y})\|^2\right] \\
=&\mathbb{E}\left[\left\|\frac{1}{M}\sum_{i=1}^{M} g_i - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y})\right\|^2\right] \\
\leq&\frac{1}{M}\mathbb{E}\left[\left\|g_i - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}, \mathbf{y})\right\|^2\right] \\
\leq&\frac{1}{M}\mathbb{E}\left[\left\|g_i\right\|^2\right] \\
\leq&\frac{16\sqrt{2\pi} d_1 L^2}{M}
\end{aligned} \tag{33}$$

where the first inequality follows because g_i are i.i.d. random variables and a sequence of i.i.d. random variable $\{\zeta_i\}_{i=1}^M$ satisfies that $\mathbb{E}\left[\left\|\frac{1}{M}\sum_{i=1}^{M}\zeta_i - \mathbb{E}[\zeta_i]\right\|^2\right] \leq \frac{1}{M}\mathbb{E}\left[\|\zeta_1 - \mathbb{E}[\zeta_1]\|^2\right]$. The second inequality follows from $\mathbb{E}\left[\|\zeta_1 - \mathbb{E}[\zeta_1]\|^2\right] \leq \mathbb{E}\left[\|\zeta_1\|^2\right]$ for any random variable ζ. The last inequality is due to Lemma D.1 in [20]. □

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B.5 Proof of Lemma 5.5

Before we give the proof of Lemma 5.5, we first give the following useful lemma.

Lemma B.1. *Let* $\mathbf{y}^*(\mathbf{x}) = \arg\max_{\mathbf{y}\in\mathcal{Y}} f(\mathbf{x},\mathbf{y})$*, we have* $\mathbf{y}^*_\delta(\mathbf{x}) = \mathbf{y}^*(\mathbf{x}+\delta\mathbf{u}) = \arg\max_{\mathbf{y}\in\mathcal{Y}} \mathbb{E}[f(\mathbf{x}+\delta\mathbf{u},\mathbf{y})]$*. Under Assumptions 3.1 and 5.1, we can obtain* $\mathbf{y}^*(\mathbf{x})$ *is* κ*-Lipschitz, where* $\kappa = \frac{L_{fy}}{\tau}$*.*

Proof. Since $f(\mathbf{x},\mathbf{y})$ is strongly concave in $\mathbf{y}$ for each $\mathbf{x}\in\mathbb{R}^{d_1}$, a function $\mathbf{y}^*(\mathbf{x})$ is unique and well-defined. Let $\mathbf{x}_1,\mathbf{x}_2\in\mathbb{R}^{d_1}$, the optimality of $\mathbf{y}^*(\mathbf{x}_1),\mathbf{y}^*(\mathbf{x}_2)$ implies that

$$(\mathbf{y}-\mathbf{y}^*(\mathbf{x}_1))^\top \nabla_{\mathbf{y}} f(\mathbf{x}_1,\mathbf{y}^*(\mathbf{x}_1)) \le 0,\ \forall \mathbf{y}\in\mathcal{Y}, \tag{34}$$

$$(\mathbf{y}-\mathbf{y}^*(\mathbf{x}_2))^\top \nabla_{\mathbf{y}} f(\mathbf{x}_2,\mathbf{y}^*(\mathbf{x}_2)) \le 0,\ \forall \mathbf{y}\in\mathcal{Y}. \tag{35}$$

Let $\mathbf{y}=\mathbf{y}^*(\mathbf{x}_2)$ in Eqn. (34) and $\mathbf{y}=\mathbf{y}^*(\mathbf{x}_1)$ in Eqn. (35). Summing the resulting two inequalities yields

$$(\mathbf{y}^*(\mathbf{x}_2)-\mathbf{y}^*(\mathbf{x}_1))^\top (\nabla_{\mathbf{y}} f(\mathbf{x}_1,\mathbf{y}^*(\mathbf{x}_1)) - \nabla_{\mathbf{y}} f(x_2,\mathbf{y}^*(\mathbf{x}_2))) \le 0. \tag{36}$$

Recall that $f(\mathbf{x}_1,\cdot)$ is τ-strongly concave, we have

$$(\mathbf{y}^*(\mathbf{x}_2)-\mathbf{y}^*(\mathbf{x}_1))^\top (\nabla_{\mathbf{y}} f(\mathbf{x}_1,\mathbf{y}^*(\mathbf{x}_2)) - \nabla_{\mathbf{y}} f(\mathbf{x}_1,\mathbf{y}^*(\mathbf{x}_1))) + \tau\|\mathbf{y}^*(\mathbf{x}_2)-\mathbf{y}^*(\mathbf{x}_1)\|^2 \le 0. \tag{37}$$

Then, using the above inequalities, we can obtain

$$\begin{aligned}
&\tau\|\mathbf{y}^*(\mathbf{x}_2)-\mathbf{y}^*(\mathbf{x}_1)\|^2 \\
\le&(\mathbf{y}^*(\mathbf{x}_2)-\mathbf{y}^*(\mathbf{x}_1))^\top (\nabla_{\mathbf{y}} f(x_2,\mathbf{y}^*(\mathbf{x}_2)) - \nabla_{\mathbf{y}} f(\mathbf{x}_1,\mathbf{y}^*(\mathbf{x}_2))) \\
\le&L_{fy}\|\mathbf{y}^*(\mathbf{x}_2)-\mathbf{y}^*(\mathbf{x}_1)\|\|\mathbf{x}_2-\mathbf{x}_1\|
\end{aligned} \tag{38}$$

Therefore, we have

$$\|\mathbf{y}^*(\mathbf{x}_2)-\mathbf{y}^*(\mathbf{x}_1)\| \le \frac{L_{fy}}{\tau}\|\mathbf{x}_2-\mathbf{x}_1\| \tag{39}$$

This completes the proof. □

Then, based on the above Lemma, we give the detailed proof of Lemma 5.5.

Proof. Since $\mathbf{y}^*_\delta(\mathbf{x})$ is unique and $\mathcal{Y}$ is convex and bounded, we conclude from Danskin's theorem that Φ is differentiable with $\nabla\Phi(x) = \nabla_{\mathbf{x}} f_\delta(\mathbf{x},\mathbf{y}^*_\delta(\mathbf{x}))$. Since $\nabla\Phi(x) = \nabla_{\mathbf{x}} f_\delta(\mathbf{x},\mathbf{y}^*_\delta(\mathbf{x}))$, we have

$$\begin{aligned}
&\|\nabla\Phi(x)-\nabla\Phi(x')\| \\
=&\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x},\mathbf{y}^*_\delta(\mathbf{x})) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}',\mathbf{y}^*_\delta(\mathbf{x}'))\| \\
\le&\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x},\mathbf{y}^*_\delta(\mathbf{x})) - \nabla_{\mathbf{x}} f_\delta(x',\mathbf{y}^*_\delta(\mathbf{x}))\| + \|\nabla_{\mathbf{x}} f_\delta(x',\mathbf{y}^*_\delta(\mathbf{x})) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}',\mathbf{y}^*_\delta(\mathbf{x}'))\| \\
\le&\frac{cL\sqrt{d_1}}{\delta}\|\mathbf{x}-\mathbf{x}'\| + L_{fxy}\|\mathbf{y}^*_\delta(\mathbf{x})-\mathbf{y}^*_\delta(\mathbf{x}')\| \\
=&\frac{cL\sqrt{d_1}}{\delta}\|\mathbf{x}-\mathbf{x}'\| + L_{fxy}\|y^*(\mathbf{x}+\delta\mathbf{u})-\mathbf{y}^*(\mathbf{x}'+\delta\mathbf{u})\| \\
\le&(\frac{cL\sqrt{d_1}}{\delta}+L_{fxy}\kappa)\|\mathbf{x}-\mathbf{x}'\|
\end{aligned} \tag{40}$$

where $\kappa = \frac{L_{fy}}{\tau}$.

This completes the proof. □

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B.6 Proof of Proposition 5.7

Proof. Assume a point $\hat{\mathbf{x}}$ statisfies that $\|\nabla\Phi(\hat{\mathbf{x}})\| \leq \epsilon$, the optimization problem $\max_{\mathbf{y}\in\mathcal{Y}} f_\delta(\hat{\mathbf{x}}, \mathbf{y})$ is strongly concave and $\mathbf{y}^*(\hat{\mathbf{x}})$ is uniquely defined. We apply the gradient descent for solving such a problem and obtain a point $\mathbf{y}' \in \mathcal{Y}$ satisfying that

$$\left\|\mathcal{P}_\mathcal{Y}\left(\mathbf{y}' + \frac{1}{L_{fy}}\nabla_\mathbf{y} f_\delta(\hat{\mathbf{x}}, \mathbf{y}')\right) - \mathbf{y}'\right\| \leq \frac{\epsilon}{L_{fy}},\ \|\mathbf{y}' - \mathbf{y}_\delta^*(\hat{\mathbf{x}})\| \leq \epsilon \tag{41}$$

We have

$$\begin{aligned}
&\left\|\mathcal{P}_\mathcal{Y}\left(\mathbf{y}' + \frac{1}{L_{fy}}\nabla_\mathbf{y} f(\hat{\mathbf{x}}, \mathbf{y}')\right) - \mathbf{y}'\right\| \\
\leq&\left\|\mathcal{P}_\mathcal{Y}\left(\mathbf{y}' + \frac{1}{L_{fy}}\nabla_\mathbf{y} f(\hat{\mathbf{x}}, \mathbf{y}')\right) - \mathcal{P}_\mathcal{Y}\left(\mathbf{y}' + \frac{1}{L_{fy}}\nabla_\mathbf{y} f_\delta(\hat{\mathbf{x}}, \mathbf{y}')\right)\right\| + \left\|\mathcal{P}_\mathcal{Y}\left(\mathbf{y}' + \frac{1}{L_{fy}}\nabla_\mathbf{y} f_\delta(\hat{\mathbf{x}}, \mathbf{y}')\right) - \mathbf{y}'\right\| \\
\leq&\frac{1}{L_{fy}}\|\nabla_\mathbf{y} f(\hat{\mathbf{x}}, \mathbf{y}') - \nabla_\mathbf{y} f_\delta(\hat{\mathbf{x}}, \mathbf{y}')\| + \frac{\epsilon}{L_{fy}} \\
=&\frac{1}{L_{fy}}\|\nabla_\mathbf{y} f(\hat{\mathbf{x}}, \mathbf{y}') - \mathbb{E}[\nabla_\mathbf{y} f(\hat{\mathbf{x}} + \delta\mathbf{u}, \mathbf{y}')]\| + \frac{\epsilon}{L_{fy}} \\
\leq&\frac{1}{L_{fy}}\mathbb{E}[\|\nabla_\mathbf{y} f(\hat{\mathbf{x}}, \mathbf{y}') - \nabla_\mathbf{y} f(\hat{\mathbf{x}} + \delta\mathbf{u}, \mathbf{y}')\|] + \frac{\epsilon}{L_{fy}} \\
\leq&\delta + \frac{\epsilon}{L_{fy}} \\
=&\mathcal{O}(\epsilon)
\end{aligned} \tag{42}$$

where the second inequality is due to the non-expansive projection operator [27, 31] and the last equality is due to $\delta = \mathcal{O}(\epsilon)$.

If $\|\nabla\Phi(\hat{\mathbf{x}})\| \leq \epsilon$, we have

$$\begin{aligned}
&\|\nabla_\mathbf{x} f_\delta(\hat{\mathbf{x}}, \mathbf{y}')\| \\
\leq&\|\nabla_\mathbf{x} f_\delta(\hat{\mathbf{x}}, \mathbf{y}') - \nabla\Phi(\hat{\mathbf{x}})\| + \|\nabla\Phi(\hat{\mathbf{x}})\| \\
\leq&\|\nabla_\mathbf{x} f_\delta(\hat{\mathbf{x}}, \mathbf{y}') - \nabla_\mathbf{x} f_\delta(\hat{\mathbf{x}}, \mathbf{y}_\delta^*(\mathbf{x}))\| + \epsilon \\
\leq&L_{fxy}\|\mathbf{y}' - \mathbf{y}_\delta^*(\hat{\mathbf{x}})\| + \epsilon \\
\leq&\mathcal{O}(\epsilon)
\end{aligned} \tag{43}$$

Conversely, if a point $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ satisfies $\|\nabla_\mathbf{x} f_\delta(\hat{\mathbf{x}}, \hat{\mathbf{y}})\| \leq \epsilon$ and $\|\mathcal{P}_\mathcal{Y}(\hat{y} + \frac{1}{L_{fy}}\nabla_\mathbf{y} f(\hat{\mathbf{x}}, \hat{\mathbf{y}})) - \hat{\mathbf{y}}\| \leq \frac{\epsilon}{L_{fy}}$, then we have

$$\begin{aligned}
&\|\nabla\Phi(x)\| \\
\leq&\|\nabla\Phi(x) - \nabla_\mathbf{x} f_\delta(\hat{\mathbf{x}}, \hat{\mathbf{y}})\| + \|\nabla_\mathbf{x} f_\delta(\hat{\mathbf{x}}, \hat{\mathbf{y}})\| \\
\leq&L_{fxy}\|\mathbf{y}_\delta^*(\hat{\mathbf{x}}) - \hat{\mathbf{y}}\| + \epsilon \\
\leq&L_{fxy}\|\mathbf{y}_\delta^*(\hat{\mathbf{x}}) - \mathbf{y}^*(\hat{\mathbf{x}})\| + L_{fxy}\|\mathbf{y}^*(\hat{\mathbf{x}}) - \hat{\mathbf{y}}\| + \epsilon \\
=&L_{fxy}\|\mathbf{y}^*(\hat{\mathbf{x}} + \delta\mathbf{u}) - \mathbf{y}^*(\hat{\mathbf{x}})\| + L_{fxy}\|\mathbf{y}^*(\hat{\mathbf{x}}) - \hat{\mathbf{y}}\| + \epsilon \\
\leq&L_{fxy}\kappa\|\hat{\mathbf{x}} + \delta\mathbf{u} - \hat{\mathbf{x}}\| + L_{fxy}\|\mathbf{y}^*(\mathbf{x}) - \hat{\mathbf{y}}\| + \epsilon \\
\leq&L_{fxy}\kappa\delta + \epsilon + L_{fxy}\|\mathbf{y}^*(\mathbf{x}) - \hat{\mathbf{y}}\|
\end{aligned} \tag{44}$$

Since $f(\hat{\mathbf{x}}, \cdot)$ is τ-strongly-concave over $\mathcal{Y}$, the global error bound condition [5] holds and

$$\tau\|\hat{\mathbf{y}} - \mathbf{y}^*(\hat{\mathbf{x}})\| \leq L_{fy}\|\mathcal{P}_\mathcal{Y}(\hat{\mathbf{y}} + \frac{1}{L_{fy}}\nabla_\mathbf{y} f(\hat{\mathbf{x}}, \hat{\mathbf{y}})) - \hat{\mathbf{y}}\| \leq \epsilon \tag{45}$$

Therefore, we conclude that

$$\|\nabla\Phi(\mathbf{x})\| \leq L_{fxy}\kappa\delta + \epsilon + \epsilon\kappa = \mathcal{O}(\epsilon) \tag{46}$$

where the last equality is due to $\delta = \mathcal{O}(\epsilon)$.

This completes the proof. □

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B.7 Detailed Proof of Convergence Analysis

In this subsection, we present several important lemmas used in our analysis and then provide the proof of our convergence result.

Lemma B.2. *When the gradient estimator $\mathbf{w}_t$ generated from Algorithm 1, for all $t \geq 1$, we have*

$$\|\mathbf{w}_t - \nabla\Phi(\mathbf{x}_t)\|^2 \leq 2\|\mathbf{w}_t - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t)\|^2 + L_0^2\kappa^2\delta^2 + L_0^2\|\mathbf{y}^*(\mathbf{x}_t) - \mathbf{y}_t\|^2 \tag{47}$$

where $L_0^2 = 4L_{fxy}^2$.

Proof. Since $\nabla\Phi(\mathbf{x}_t) = \nabla_{\mathbf{x}} f(\mathbf{x}_t, \mathbf{y}_\delta^*(\mathbf{x}_t))$, we have

$$\begin{aligned}
&\|\nabla\Phi(\mathbf{x}_t) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t)\|^2 \\
=&\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_\delta^*(\mathbf{x}_t)) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t)\|^2 \\
\leq& L_{fxy}^2\|\mathbf{y}_\delta^*(\mathbf{x}_t) - \mathbf{y}_t\|^2 \\
\leq& 2L_{fxy}^2\|\mathbf{y}_\delta^*(\mathbf{x}_t) - \mathbf{y}^*(\mathbf{x}_t)\|^2 + 2L_{fxy}^2\|\mathbf{y}^*(\mathbf{x}_t) - \mathbf{y}_t\|^2 \\
=& 2L_{fxy}^2\|\mathbf{y}^*(\mathbf{x}_t + \delta\mathbf{u}) - \mathbf{y}^*(\mathbf{x}_t)\|^2 + 2L_{fxy}^2\|\mathbf{y}^*(\mathbf{x}_t) - \mathbf{y}_t\|^2 \\
\leq& 2L_{fxy}^2\kappa^2\delta^2 + 2L_{fxy}^2\|\mathbf{y}^*(\mathbf{x}_t) - \mathbf{y}_t\|^2
\end{aligned} \tag{48}$$

Then, we have

$$\begin{aligned}
&\|\mathbf{w}_t - \nabla\Phi(\mathbf{x}_t)\|^2 \\
\leq& 2\|\mathbf{w}_t - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t)\|^2 + 2\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t) - \nabla\Phi(\mathbf{x}_t)\|^2 \\
\leq& 2\|\mathbf{w}_t - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t)\|^2 + 4L_{fxy}^2\kappa^2\delta^2 + 4L_{fxy}^2\|\mathbf{y}^*(\mathbf{x}_t) - \mathbf{y}_t\|^2
\end{aligned} \tag{49}$$

□

Lemma B.3. *Suppose that the sequence $\{\mathbf{x}_t, \mathbf{y}_t\}_{t=1}^T$ be generated from Algorithm 1, let $0 < \eta_t \leq 1$ and $0 < \gamma \leq \frac{1}{2\ell\eta_t}$, then we have*

$$\Phi(\mathbf{x}_{t+1}) \leq \Phi(\mathbf{x}_t) + \gamma\eta_t\|\nabla\Phi(\mathbf{x}_t) - \mathbf{w}_t\|^2 - \frac{\eta_t}{2\gamma}\|\mathbf{x}_t - \tilde{\mathbf{x}}_{t+1}\|^2 \tag{50}$$

Proof. According to Lemma 5.5, the function $\Phi(x)$ is ℓ-smooth. Thus we have

$$\begin{aligned}
&\Phi(\mathbf{x}_{t+1}) \\
\leq&\Phi(\mathbf{x}_t) + \langle\nabla\Phi(\mathbf{x}_t), \mathbf{x}_{t+1} - \mathbf{x}_t\rangle + \frac{\ell}{2}\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
=&\Phi(\mathbf{x}_t) + \langle\nabla\Phi(\mathbf{x}_t), \eta_t(\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t)\rangle + \frac{\ell}{2}\|\eta_t(\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t)\|^2 \\
=&\Phi(\mathbf{x}_t) + \eta_t\langle\mathbf{w}_t, \tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\rangle + \eta_t\langle\nabla\Phi(\mathbf{x}_t) - \mathbf{w}_t, \tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\rangle + \frac{\ell\eta_t^2}{2}\|\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\|^2
\end{aligned} \tag{51}$$

where the second equality is due to $\mathbf{x}_{t+1} = \mathbf{x}_t + \eta_t(\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t)$.

Since $\tilde{\mathbf{x}}_{t+1} = \mathbf{x}_t - \gamma\mathbf{w}_t = \arg\min_{x\in\mathbb{R}^{d_1}}\{\langle\mathbf{w}_t, \mathbf{x}\rangle + \frac{1}{2\gamma}\|\mathbf{x} - \mathbf{x}_t\|^2\}$, we have

$$\langle\mathbf{w}_t, \frac{1}{\gamma}(\mathbf{x}_t - \tilde{\mathbf{x}}_{t+1})\rangle \geq \|\frac{1}{\gamma}(\mathbf{x}_t - \tilde{\mathbf{x}}_{t+1})\|^2 \tag{52}$$

Then, we have

$$\langle\mathbf{w}_t, \tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\rangle \leq -\frac{1}{\gamma}\|\mathbf{x}_t - \tilde{\mathbf{x}}_{t+1}\|^2 \tag{53}$$

In addition, we have

$$\begin{aligned}
&\langle\nabla\Phi(\mathbf{x}_t) - \mathbf{w}_t, \tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\rangle \\
\leq&\|\nabla\Phi(\mathbf{x}_t) - \mathbf{w}_t\| \cdot \|\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\| \\
\leq&\gamma\|\nabla\Phi(\mathbf{x}_t) - \mathbf{w}_t\|^2 + \frac{1}{4\gamma}\|\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\|^2
\end{aligned} \tag{54}$$

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where the first inequality is due to the Cauchy-Schwarz inequality and the last is due to Young's inequality. Combining the above inequalities, we obtain

$$\begin{aligned}
&\Phi(\mathbf{x}_{t+1}) \\
=&\Phi(\mathbf{x}_t) + \eta_t\langle \mathbf{w}_t, \tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\rangle + \eta_t\langle \nabla\Phi(\mathbf{x}_t) - \mathbf{w}_t, \tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\rangle + \frac{\ell\eta_t^2}{2}\|\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\|^2 \\
\leq&\Phi(\mathbf{x}_t) - \frac{\eta_t}{\gamma}\|\mathbf{x} - \tilde{\mathbf{x}}_{t+1}\|^2 + \gamma\eta_t\|\nabla\Phi(\mathbf{x}_t) - \mathbf{w}_t\|^2 + \frac{\eta_t}{4\gamma}\|\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\|^2 + \frac{\ell\eta_t^2}{2}\|\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\|^2 \\
=&\Phi(\mathbf{x}_t) + \gamma\eta_t\|\nabla\Phi(\mathbf{x}_t) - \mathbf{w}_t\|^2 - \frac{\eta_t}{2\gamma}\|\mathbf{x} - \tilde{\mathbf{x}}_{t+1}\|^2 - (\frac{\eta_t}{4\gamma} - \frac{\ell\eta_t^2}{2})\|\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\|^2 \\
\leq&\Phi(\mathbf{x}_t) + \gamma\eta_t\|\nabla\Phi(\mathbf{x}_t) - \mathbf{w}_t\|^2 - \frac{\eta_t}{2\gamma}\|\mathbf{x}_t - \tilde{\mathbf{x}}_{t+1}\|^2
\end{aligned} \tag{55}$$

where the last inequality is due to $0 < \gamma \leq \frac{1}{2\ell\eta_t}$. □

Lemma B.4. *Suppose the sequence $\{\mathbf{x}_t, \mathbf{y}_t\}_{t=1}^T$ be generated from Algorithm 1. Given $0 < \eta_t \leq 1$, for all $t \geq 1$ and $0 < \lambda \leq \frac{1}{6L_{fy}}$, we have*

$$\begin{aligned}
&\|\mathbf{y}_{t+1} - \mathbf{y}^*(\mathbf{x}_{t+1})\|^2 \\
\leq&(1 - \frac{\eta_t\tau\lambda}{4})\|\mathbf{y}_t - \mathbf{y}^*(\mathbf{x}_t)\|^2 - \frac{3\eta_t}{4}\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2 + \frac{25\eta_t\lambda}{6\tau}\|\nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t\|^2 + \frac{25\eta_t\kappa^2}{6\tau\lambda}\|\mathbf{x}_t - \tilde{\mathbf{x}}_{t+1}\|^2
\end{aligned} \tag{56}$$

where $\kappa = \frac{L_{fy}}{\tau}$.

Proof. Since $g(\mathbf{x}, \mathbf{y}) = -f(\mathbf{x}, \mathbf{y})$ is τ-strongly convex w.r.t. y, we have

$$\begin{aligned}
&g(\mathbf{x}_t, \mathbf{y}) \\
\geq&g(\mathbf{x}_t, \mathbf{y}_t) + \langle \nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t), \mathbf{y} - \mathbf{y}_t\rangle + \frac{\tau}{2}\|\mathbf{y} - \mathbf{y}_t\|^2 \\
=&g(\mathbf{x}_t, \mathbf{y}_t) + \langle \mathbf{v}_t, \mathbf{y} - \tilde{\mathbf{y}}_{t+1}\rangle + \langle \nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t, \mathbf{y} - \tilde{\mathbf{y}}_{t+1}\rangle + \langle \nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t), \tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\rangle + \frac{\tau}{2}\|\mathbf{y} - \mathbf{y}_t\|^2
\end{aligned} \tag{57}$$

Since $g(\mathbf{x}, \mathbf{y})$ is L_{fy}-smooth w.r.t. y, we have

$$g(\mathbf{x}_t, \tilde{\mathbf{y}}_{t+1}) \leq g(\mathbf{x}_t, \mathbf{y}_t) + \langle \nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t), \tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\rangle + \frac{L_{fy}}{2}\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2 \tag{58}$$

Using the above inequality, we have

$$g(\mathbf{x}_t, \mathbf{y}) \geq g(\mathbf{x}_t, \tilde{\mathbf{y}}_{t+1}) + \langle \mathbf{v}_t, \mathbf{y} - \tilde{\mathbf{y}}_{t+1}\rangle + \langle \nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t, \mathbf{y} - \tilde{\mathbf{y}}_{t+1}\rangle + \frac{\tau}{2}\|\mathbf{y} - \mathbf{y}_t\|^2 - \frac{L_{fy}}{2}\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2 \tag{59}$$

By the optimality condition of problem $\tilde{\mathbf{y}}_{t+1} = \arg\min_{\mathbf{y}\in\mathcal{Y}}\{\langle \mathbf{v}_t, \mathbf{y}\rangle + \frac{1}{2\lambda}(\mathbf{y} - \mathbf{y}_t)^\top(\mathbf{y} - \mathbf{y}_t)\}$, we have

$$\langle \mathbf{v}_t + \frac{1}{\lambda}(\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t), \mathbf{y} - \tilde{\mathbf{y}}_{t+1}\rangle \geq 0 \tag{60}$$

Then we have

$$\langle \mathbf{v}_t, \mathbf{y} - \tilde{\mathbf{y}}_{t+1}\rangle \geq \frac{1}{\lambda}\langle \tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t, \tilde{\mathbf{y}}_{t+1} - y\rangle = \frac{1}{\lambda}\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2 + \frac{1}{\lambda}\langle \tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t, \mathbf{y}_t - \mathbf{y}\rangle \tag{61}$$

Using the above inequalities, we have

$$\begin{aligned}
g(\mathbf{x}_t, \mathbf{y}) \geq&g(\mathbf{x}_t, \tilde{\mathbf{y}}_{t+1}) + \frac{1}{\lambda}\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2 + \frac{1}{\lambda}\langle \tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t, \mathbf{y}_t - \mathbf{y}\rangle \\
&+ \langle \nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t, \mathbf{y} - \tilde{\mathbf{y}}_{t+1}\rangle + \frac{\tau}{2}\|\mathbf{y} - \mathbf{y}_t\|^2 - \frac{L_{fy}}{2}\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2
\end{aligned} \tag{62}$$

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Let $\mathbf{y} = \mathbf{y}^*(\mathbf{x}_t)$, then we have

$$\begin{aligned} g(\mathbf{x}_t, \mathbf{y}^*(\mathbf{x}_t)) \geq & g(\mathbf{x}_t, \tilde{\mathbf{y}}_{t+1}) + (\frac{1}{\lambda} - \frac{L_{fy}}{2})\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2 + \frac{1}{\lambda}\langle \tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t, \mathbf{y}_t - \mathbf{y}^*(\mathbf{x}_t)\rangle \\ & + \langle \nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t, \mathbf{y}^*(\mathbf{x}_t) - \tilde{\mathbf{y}}_{t+1}\rangle + \frac{\tau}{2}\|\mathbf{y}^*(\mathbf{x}_t) - \mathbf{y}_t\|^2 \end{aligned} \tag{63}$$

Since g is strongly convex w.r.t. $\mathbf{y}$, we have

$$\begin{aligned} 0 \geq & (\frac{1}{\lambda} - \frac{L_{fy}}{2})\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2 + \frac{1}{\lambda}\langle \tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t, \mathbf{y}_t - \mathbf{y}^*(\mathbf{x}_t)\rangle \\ & + \langle \nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t, \mathbf{y}^*(\mathbf{x}_t) - \tilde{\mathbf{y}}_{t+1}\rangle + \frac{\tau}{2}\|\mathbf{y}^*(\mathbf{x}_t) - \mathbf{y}_t\|^2 \end{aligned} \tag{64}$$

Since $\mathbf{y}_{t+1} = \mathbf{y}_t + \eta_t(\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t)$, we have

$$\begin{aligned} & \|\mathbf{y}_{t+1} - \mathbf{y}^*(\mathbf{x}_t)\|^2 \\ = & \|\mathbf{y}_t + \eta_t(\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t) - \mathbf{y}^*(\mathbf{x}_t)\|^2 \\ = & \|\mathbf{y}_t - \mathbf{y}^*(\mathbf{x}_t)\|^2 + \eta_t^2\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2 + 2\eta_t\langle \mathbf{y}_t - \mathbf{y}^*(\mathbf{x}_t), \tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\rangle \end{aligned} \tag{65}$$

Then, we have

$$\langle \mathbf{y}_t - \mathbf{y}^*(\mathbf{x}_t), \tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\rangle = \frac{1}{2\eta_t}\|\mathbf{y}_{t+1} - \mathbf{y}^*(\mathbf{x}_t)\|^2 - \frac{1}{2\eta_2}\|\mathbf{y}_t - \mathbf{y}^*(\mathbf{x}_t)\|^2 - \frac{\eta_t}{2}\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|. \tag{66}$$

Consider the upper bound of $\langle \nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t, \mathbf{y}^*(\mathbf{x}_t) - \tilde{\mathbf{y}}_{t+1}\rangle$, we have

$$\begin{aligned} & \langle \nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t, \mathbf{y}^*(\mathbf{x}_t) - \tilde{\mathbf{y}}_{t+1}\rangle \\ = & \langle \nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t, \mathbf{y}^*(\mathbf{x}_t) - y_t\rangle + \langle \nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t, y_t - \tilde{\mathbf{y}}_{t+1}\rangle \\ \geq & -\frac{1}{\tau}\|\nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t\|^2 - \frac{\tau}{4}\|\mathbf{y}^*(\mathbf{x}_t) - \mathbf{y}_t\|^2 - \frac{1}{\tau}\|\nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t\|^2 - \frac{\tau}{4}\|\mathbf{y}_t - \tilde{\mathbf{y}}_{t+1}\|^2 \\ = & -\frac{2}{\tau}\|\nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t\|^2 - \frac{\tau}{4}\|\mathbf{y}^*(\mathbf{x}_t) - \mathbf{y}_t\|^2 - \frac{\tau}{4}\|\mathbf{y}_t - \tilde{\mathbf{y}}_{t+1}\|^2 \end{aligned} \tag{67}$$

Using the above inequalities and $0 < \lambda \leq \frac{1}{6L_{Lfy}}$, we have

$$\frac{1}{2\eta_t\lambda}\|\mathbf{y}_{t+1} - \mathbf{y}^*(\mathbf{x}_t)\|^2 \leq (\frac{1}{2\eta_t\lambda} - \frac{\tau}{4})\|\mathbf{y}_t - \mathbf{y}^*(\mathbf{x}_t)\|^2 - \frac{3}{8\lambda}\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2 + \frac{2}{\tau}\|\nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t\|^2. \tag{68}$$

Therefore, we have

$$\|\mathbf{y}_{t+1} - \mathbf{y}^*(\mathbf{x}_t)\|^2 \leq (1 - \frac{\tau\eta_t\lambda}{2})\|\mathbf{y}_t - \mathbf{y}^*(\mathbf{x}_t)\|^2 - \frac{3\eta_t}{4}\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2 + \frac{4\eta_t\lambda}{\tau}\|\nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t\|^2. \tag{69}$$

In addition, we have

$$\begin{aligned} & \|\mathbf{y}_{t+1} - \mathbf{y}^*(\mathbf{x}_{t+1})\|^2 \\ = & \|\mathbf{y}_{t+1} - \mathbf{y}^*(\mathbf{x}_t) + \mathbf{y}^*(\mathbf{x}_t) - \mathbf{y}^*(\mathbf{x}_{t+1})\|^2 \\ = & \|\mathbf{y}_{t+1} - \mathbf{y}^*(\mathbf{x}_t)\|^2 + 2\langle \mathbf{y}_{t+1} - \mathbf{y}^*(\mathbf{x}_t), \mathbf{y}^*(\mathbf{x}_t) - \mathbf{y}^*(\mathbf{x}_{t+1})\rangle + \|\mathbf{y}^*(\mathbf{x}_t) - \mathbf{y}^*(\mathbf{x}_{t+1})\|^2 \\ \leq & (1 + \frac{\eta_t\tau\lambda}{4})\|\mathbf{y}_{t+1} - \mathbf{y}^*(\mathbf{x}_t)\|^2 + (1 + \frac{4}{\eta_t\tau\lambda})\|\mathbf{y}^*(\mathbf{x}_t) - \mathbf{y}^*(\mathbf{x}_{t+1})\|^2 \\ \leq & (1 + \frac{\eta_t\tau\lambda}{4})\|\mathbf{y}_{t+1} - \mathbf{y}^*(\mathbf{x}_t)\|^2 + (1 + \frac{4}{\eta_t\tau\lambda})\kappa^2\|\mathbf{x}_t - \mathbf{x}_{t+1}\|^2 \end{aligned} \tag{70}$$

where the last inequality is due to Lemma B.1.

Usin the above inequalities, we have

$$\begin{aligned} \|\mathbf{y}_{t+1} - \mathbf{y}^*(\mathbf{x}_{t+1})\|^2 \leq & (1 + \frac{\eta_t\tau\lambda}{4})(1 - \frac{\tau\eta_t\lambda}{2})\|\mathbf{y}_t - \mathbf{y}^*(\mathbf{x}_t)\|^2 - (1 + \frac{\eta_t\tau\lambda}{4})\frac{3\eta_t}{4}\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2 \\ & + (1 + \frac{\eta_t\tau\lambda}{4})\frac{4\eta_t\lambda}{\tau}\|\nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t\|^2 + (1 + \frac{4}{\eta_t\tau\lambda})\kappa^2\|\mathbf{x}_t - \mathbf{x}_{t+1}\|^2 \end{aligned} \tag{71}$$

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Since $0 < \eta_t \leq 1$, $0 < \lambda \leq \frac{1}{6L_{fy}}$ and $L_{fy} > \tau$, we have $\lambda \leq \frac{1}{6L_g} \leq \frac{1}{6\tau}$ and $\eta_t \leq 1 \leq \frac{1}{6\tau\lambda}$. Then, we have

$$(1 + \frac{\eta_t\tau\lambda}{4})(1 - \frac{\eta_t\tau\lambda}{2}) \leq 1 - \frac{\eta_t\tau\lambda}{4} \tag{72}$$

$$-(1 + \frac{\eta_t\tau\lambda}{4})\frac{3\eta_t}{4} \leq -\frac{3\eta_t}{4}, \tag{73}$$

$$(1 + \frac{\eta_t\tau\lambda}{4})\frac{4\eta_t\lambda}{\tau} \leq \frac{25\eta_t\lambda}{6\tau}, \tag{74}$$

$$(1 + \frac{4}{\eta_t\tau\lambda})\kappa^2 \leq \frac{25\kappa^2}{6\eta_t\tau\lambda}. \tag{75}$$

By using $x_{t+1} = x_t + \eta_t(\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t)$, we have

$$\begin{aligned}
&\|\mathbf{y}_{t+1} - \mathbf{y}^*(\mathbf{x}_{t+1})\|^2 \\
\leq&(1 - \frac{\eta_t\tau\lambda}{4})\|\mathbf{y}_t - \mathbf{y}^*(\mathbf{x}_t)\|^2 - \frac{3\eta_t}{4}\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2 + \frac{25\eta_t\lambda}{6\tau}\|\nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{v}_t\|^2 + \frac{25\eta_t\kappa^2}{6\tau\lambda}\|\mathbf{x}_t - \tilde{\mathbf{x}}_{t+1}\|^2
\end{aligned} \tag{76}$$

□

Lemma B.5. *Assume that the stochastic partial derivatives $\mathbf{v}_{t+1}$ and $\mathbf{w}_{t+1}$ be generated from Algorithm 1, we have*

$$\begin{aligned}
&\mathbb{E}[\|\mathbf{w}_{t+1} - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1})\|^2] \\
\leq&(1 - \beta_{t+1})\mathbb{E}[\|\mathbf{w}_t - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t)\|^2] + \beta_{t+1}^2\sigma_{\mathbf{x}}^2 + \frac{2L_{fxy}^2\eta_t^2}{\beta_{t+1}}\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2 + \frac{2c^2L^2d_1\eta_t^2}{\delta^2\beta_{t+1}}\|\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\|^2
\end{aligned} \tag{77}$$

$$\begin{aligned}
&\mathbb{E}[\|\mathbf{v}_{t+1} - \nabla_{\mathbf{y}} g(\mathbf{x}_{t+1}, \mathbf{y}_{t+1})\|^2] \\
\leq&(1 - \alpha_{t+1})\mathbb{E}[\|\mathbf{v}_t - \nabla_{\mathbf{y}} g(\mathbf{x}_t, \mathbf{y}_t)\|^2] + \frac{\alpha_{t+1}^2\sigma_{\mathbf{y}}^2}{M} + \frac{2L_{fy}^2}{\alpha_{t+1}}\eta_t^2(\|\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\|^2 + \|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2)
\end{aligned} \tag{78}$$

Proof. Since $\mathbf{w}_{t+1} = (1 - \beta_{t+1})\mathbf{w}_t + \beta_{t+1}\bar{\nabla}_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1}; \mathcal{B}_{t+1})$, we have

$$\begin{aligned}
&\mathbb{E}[\|\mathbf{w}_{t+1} - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1})\|^2] \\
=&\mathbb{E}[\|(1 - \beta_{t+1})\mathbf{w}_t + \beta_{t+1}\bar{\nabla}_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1}; \mathcal{B}_{t+1}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1})\|^2] \\
=&\mathbb{E}[\|(1 - \beta_{t+1})(\mathbf{w}_t - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t)) + \beta_{t+1}(\bar{\nabla}_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1}; \mathcal{B}_{t+1}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1})) \\
&+ (1 - \beta_{t+1})(\nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1}))\|^2] \\
=&\beta_{t+1}^2\mathbb{E}[\|\bar{\nabla}_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1}; \mathcal{B}_{t+1}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1})\|^2] + \mathbb{E}[\|(1 - \beta_{t+1})(\mathbf{w}_t - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t)) + \\
&+ (1 - \beta_{t+1})(\nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1}))\|^2] \\
\leq&\beta_{t+1}^2\mathbb{E}[\|\bar{\nabla}_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1}; \mathcal{B}_{t+1}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1})\|^2] + (1 - \beta_{t+1})^2(1 + \beta_{t+1})\mathbb{E}[\|\mathbf{w}_t - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t)\|^2] \\
&+ (1 - \beta_{t+1})^2(1 + \frac{1}{\beta_{t+1}})\mathbb{E}[\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1})\|^2] \\
\leq&\beta_{t+1}^2\sigma_{\mathbf{x}}^2 + (1 - \beta_{t+1})\mathbb{E}[\|\mathbf{w}_t - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t)\|^2] + \frac{1}{\beta_{t+1}}\mathbb{E}[\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1})\|^2] \\
\leq&(1 - \beta_{t+1})\mathbb{E}[\|\mathbf{w}_t - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t)\|^2] + \beta_{t+1}^2\sigma_{\mathbf{x}}^2 + \frac{2}{\beta_{t+1}}\mathbb{E}[\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_{t+1})\|^2] \\
&+ \frac{2}{\beta_{t+1}}\mathbb{E}[\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_{t+1}) - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_{t+1}, \mathbf{y}_{t+1})\|^2] \\
\leq&(1 - \beta_{t+1})\mathbb{E}[\|\mathbf{w}_t - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t)\|^2] + \beta_{t+1}^2\sigma_{\mathbf{x}}^2 + \frac{2}{\beta_{t+1}}L_{fxy}^2\|\mathbf{y}_{t+1} - \mathbf{y}_t\|^2 + \frac{2}{\beta_{t+1}}\frac{c^2L^2d_1}{\delta^2}\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
\leq&(1 - \beta_{t+1})\mathbb{E}[\|\mathbf{w}_t - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t)\|^2] + \beta_{t+1}^2\sigma_{\mathbf{x}}^2 + \frac{2L_{fxy}^2\eta_t^2}{\beta_{t+1}}\|\tilde{\mathbf{y}}_{t+1} - \mathbf{y}_t\|^2 + \frac{2c^2L^2d_1\eta_t^2}{\delta^2\beta_{t+1}}\|\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\|^2
\end{aligned} \tag{79}$$

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Similarly, we have

$$\begin{aligned}&\mathbb{E}[\|\mathbf{v}_{t+1}-\nabla_{\mathbf{y}}g(\mathbf{x}_{t+1},\mathbf{y}_{t+1})\|^2]\\ \leq&(1-\alpha_{t+1})\mathbb{E}[\|\mathbf{v}_t-\nabla_{\mathbf{y}}g(\mathbf{x}_t,\mathbf{y}_t)\|^2]+\frac{\alpha_{t+1}^2\sigma_{\mathbf{y}}^2}{M}+\frac{2L_{fy}^2}{\alpha_{t+1}}\eta_t^2(\|\tilde{\mathbf{x}}_{t+1}-\mathbf{x}_t\|^2+\|\tilde{\mathbf{y}}_{t+1}-\mathbf{y}_t\|^2)\end{aligned} \tag{80}$$

□

B.8 Detialed Proof of Theorem 5.10

Theorem B.6. *(Restatement of Theorem 5.10) Under Assumptions 3.1, 5.1, 5.2, and 5.9, in Algorithm 1,* $\eta_t=\frac{k}{(m+t)^{1/2}}$ *for all* $t\geq 1$, $\alpha_{t+1}=c_1\eta_t$, $\beta_{t+1}=c_2\eta_t$, $m\geq\max\{k^2,(c_1k)^2,(c_2k)^2\}$, $k>0$, $\frac{125L_0^2}{6\sqrt{\delta}\tau^2}\leq c_1\leq\frac{m^{1/2}}{k}$, $\frac{5}{2\sqrt{\delta}}\leq c_2\leq\frac{m^{1/2}}{k}$, $0<\gamma\leq\min\{\frac{1}{2\sqrt{\frac{125L_0^2\kappa^2}{6\tau^2\lambda^2}+\frac{4c^2L^2d_1}{5\delta}+\frac{12\tau^2L_{fy}^2}{125L_0^2}}},\frac{m^{1/2}}{4\ell k}\}$ *and* $0<\lambda\leq\min\{\frac{15L_0^2}{4\tau\left(\frac{4L_{fxy}^2}{5}+\frac{12\tau^2L_{fy}^2}{125L_0^2}\right)},\frac{1}{6L_{fy}}\}$, *we have*

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\|\nabla\Phi(\mathbf{x}_t)\|]\leq G', \tag{81}$$

where $G'=\frac{4\sqrt{G}m^{1/4}}{T^{1/2}}+\frac{4\sqrt{G}}{T^{1/4}}+\frac{2\sqrt{5m}L_0\kappa\delta}{T^{1/2}}+2\sqrt{5}L_0\kappa\delta$, $G=\frac{(\Phi(x_1)-\Phi^*)}{k\gamma}+\frac{5b_1L_0^2}{\lambda\tau k}\Delta_1+\frac{\sqrt{\delta}\left(16\sqrt{2\pi}d_1L^2+\sigma_y^2\right)}{Mk}+\frac{m\sqrt{\delta}\left(16\sqrt{2\pi}d_1L^2+\sigma_y^2\right)}{Mk}\ln(m+T)$, $L_0=4L_{fxy}$ *and* $\Delta_1=\|y_1-y^*(x_1)\|^2$.

Proof. Since $\eta_t=\frac{k}{(m+t)^{1/2}}$ on t is decreasing and $m\geq k^2$, we have $\eta_t\leq\eta_0=\frac{k}{m^{1/2}}\leq 1$ and $\gamma\leq\frac{m^{1/2}}{4\ell k}\leq\frac{1}{2\ell\eta_0}\leq\frac{1}{2\ell\eta_t}$. Due to $0<\eta_t\leq 1$ and $m\geq(c_1k)^2$, we have $\alpha_{t+1}=c_1\eta_t\leq\frac{c_1k}{m^{1/2}}\leq 1$. Similarly, we have $\beta_{t+1}\leq 1$. Based on our above Lemmas, let $\beta_{t+1}=c_2\eta_t$ and $\frac{5}{2\sqrt{\delta}}\leq c_2\leq\frac{m^{1/2}}{k}$, we have

$$\begin{aligned}&\mathbb{E}[\|\mathbf{w}_{t+1}-\nabla_{\mathbf{x}}f_\delta(\mathbf{x}_{t+1},\mathbf{y}_{t+1})\|^2]-\mathbb{E}[\|\mathbf{w}_t-\nabla_{\mathbf{x}}f_\delta(\mathbf{x}_t,\mathbf{y}_t)\|^2]\\ \leq&-\beta_{t+1}\mathbb{E}[\|\mathbf{w}_t-\nabla_{\mathbf{x}}f_\delta(\mathbf{x}_t,\mathbf{y}_t)\|^2]+\beta_{t+1}^2\sigma_{\mathbf{x}}^2+\frac{2L_{fxy}^2\eta_t^2}{\beta_{t+1}}\|\tilde{\mathbf{y}}_{t+1}-\mathbf{y}_t\|^2+\frac{2c^2L^2d_1\eta_t^2}{\delta^2\beta_{t+1}}\|\tilde{\mathbf{x}}_{t+1}-\mathbf{x}_t\|^2\\ \leq&-\frac{5\eta_t}{2\sqrt{\delta}}\mathbb{E}[\|\mathbf{w}_t-\nabla_{\mathbf{x}}f_\delta(\mathbf{x}_t,\mathbf{y}_t)\|^2]+\frac{m\eta_t^2\sigma_{\mathbf{x}}^2}{k^2}+\frac{4\sqrt{\delta}L_{fxy}^2\eta_t}{5}\|\tilde{\mathbf{y}}_{t+1}-\mathbf{y}_t\|^2+\frac{4\sqrt{\delta}c^2L^2d_1\eta_t}{5\delta^2}\|\tilde{\mathbf{x}}_{t+1}-\mathbf{x}_t\|^2\end{aligned} \tag{82}$$

Let $\alpha_{t+1}=c_1\eta_t$ and $\frac{125L_0^2}{6\sqrt{\delta}\tau^2}\leq c_1\leq\frac{m^{1/2}}{k}$, we have

$$\begin{aligned}&\mathbb{E}[\|\mathbf{v}_{t+1}-\nabla_{\mathbf{y}}g(\mathbf{x}_{t+1},\mathbf{y}_{t+1})\|^2]-\mathbb{E}[\|\mathbf{v}_t-\nabla_{\mathbf{y}}g(\mathbf{x}_t,\mathbf{y}_t)\|^2]\\ \leq&-c_1\eta_t\mathbb{E}[\|\mathbf{v}_t-\nabla_{\mathbf{y}}g(\mathbf{x}_t,\mathbf{y}_t)\|^2]+\alpha_{t+1}^2\sigma_{\mathbf{y}}^2+\frac{2L_{fy}^2}{\alpha_{t+1}}\eta_t^2(\|\tilde{\mathbf{x}}_{t+1}-\mathbf{x}_t\|^2+\|\tilde{\mathbf{y}}_{t+1}-\mathbf{y}_t\|^2)\\ \leq&-\frac{125L_0^2}{6\sqrt{\delta}\tau^2}\eta_t\mathbb{E}[\|\mathbf{v}_t-\nabla_{\mathbf{y}}g(\mathbf{x}_t,\mathbf{y}_t)\|^2]+\frac{m\sigma_{\mathbf{y}}^2\eta_t^2}{k^2M}+\frac{12\sqrt{\delta}\tau^2L_{fy}^2\eta_t}{125L_0^2}(\|\tilde{\mathbf{x}}_{t+1}-\mathbf{x}_t\|^2+\|\tilde{\mathbf{y}}_{t+1}-\mathbf{y}_t\|^2)\end{aligned} \tag{83}$$

We also have

$$\begin{aligned}&\Phi(\mathbf{x}_{t+1})-\Phi(\mathbf{x}_t)\\ \leq&\gamma\eta_t\|\nabla\Phi(\mathbf{x}_t)-\mathbf{w}_t\|^2-\frac{\eta_t}{2\gamma}\|\mathbf{x}-\tilde{\mathbf{x}}_{t+1}\|^2\\ \leq&2\gamma\eta_t\|\mathbf{w}_t-\nabla_{\mathbf{x}}f_\delta(\mathbf{x}_t,\mathbf{y}_t)\|^2+\gamma\eta_tL_0^2\kappa^2\delta^2+L_0^2\gamma\eta_t\|\mathbf{y}^*(\mathbf{x}_t)-\mathbf{y}_t\|^2-\frac{\eta_t}{2\gamma}\|\mathbf{x}_t-\tilde{\mathbf{x}}_{t+1}\|^2\end{aligned} \tag{84}$$

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We have

$$\begin{aligned}
&\|\mathbf{y}_{t+1}-\mathbf{y}^*(\mathbf{x}_{t+1})\|^2-\|\mathbf{y}_t-\mathbf{y}^*(\mathbf{x}_t)\|^2\\
\leq&-\frac{\eta_t\tau\lambda}{4}\|\mathbf{y}_t-\mathbf{y}^*(\mathbf{x}_t)\|^2-\frac{3\eta_t}{4}\|\tilde{\mathbf{y}}_{t+1}-\mathbf{y}_t\|^2+\frac{25\eta_t\lambda}{6\tau}\|\nabla_{\mathbf{y}}g(\mathbf{x}_t,\mathbf{y}_t)-\mathbf{v}_t\|^2+\frac{25\eta_t\kappa^2}{6\tau\lambda}\|\mathbf{x}_t-\tilde{\mathbf{x}}_{t+1}\|^2
\end{aligned}\tag{85}$$

Define a Lyapunov function, for any $t\geq 1$, we have

$$\Gamma_t=\mathbb{E}[\Phi(\mathbf{x}_t)+\frac{5L_0^2\gamma}{\lambda\tau}\|\mathbf{y}_t-\mathbf{y}^*(\mathbf{x}_t)\|^2+\gamma\sqrt{\delta}\|\mathbf{v}_t-\nabla_{\mathbf{y}}g(\mathbf{x}_t,\mathbf{y}_t)\|^2+\gamma\sqrt{\delta}\|\mathbf{w}_t-\nabla_{\mathbf{x}}f_\delta(\mathbf{x}_t,\mathbf{y}_t)\|^2]\tag{86}$$

Then we have

$$\begin{aligned}
&\Gamma_{t+1}-\Gamma_t\\
=&\mathbb{E}[\Phi(\mathbf{x}_{t+1})-\Phi(\mathbf{x}_t)+\frac{5L_0^2\gamma}{\lambda\tau}(\|\mathbf{y}_{t+1}-\mathbf{y}^*(\mathbf{x}_{t+1})\|^2-\|\mathbf{y}_t-\mathbf{y}^*(\mathbf{x}_t)\|^2)\\
&+\gamma(\|\mathbf{v}_{t+1}-\nabla_{\mathbf{y}}g(\mathbf{x}_{t+1},\mathbf{y}_{t+1})\|^2-\|\mathbf{v}_t-\nabla_{\mathbf{y}}g(\mathbf{x}_t,\mathbf{y}_t)\|^2)+\gamma\sqrt{\delta}(\|\mathbf{w}_{t+1}-\nabla_{\mathbf{x}}f_\delta(\mathbf{x}_{t+1},\mathbf{y}_{t+1})\|^2-\|\mathbf{w}_t-\nabla_{\mathbf{x}}f_\delta(\mathbf{x}_t,\mathbf{y}_t)\|\\
\leq&\frac{2\gamma\eta_t}{}\mathbb{E}[\|\mathbf{w}_t-\nabla_{\mathbf{x}}f_\delta(\mathbf{x}_t,\mathbf{y}_t)\|^2]+\gamma\eta_tL_0^2\kappa^2\delta^2+\frac{L_0^2\gamma\eta_t}{}\mathbb{E}[\|\mathbf{y}^*(\mathbf{x}_t)-\mathbf{y}_t\|^2]-\frac{\eta_t}{2\gamma}\mathbb{E}[\|\mathbf{x}_t-\tilde{\mathbf{x}}_{t+1}\|^2]\\
&+\frac{5L_0^2\gamma}{\lambda\tau}\left(-\frac{\eta_t\tau\lambda}{4}\mathbb{E}[\|\mathbf{y}_t-\mathbf{y}^*(\mathbf{x}_t)\|^2]-\frac{3\eta_t}{4}\mathbb{E}[\|\tilde{\mathbf{y}}_{t+1}-\mathbf{y}_t\|^2]+\frac{25\eta_t\lambda}{6\tau}\mathbb{E}[\|\nabla_{\mathbf{y}}g(\mathbf{x}_t,\mathbf{y}_t)-\mathbf{v}_t\|^2]+\frac{25\eta_t\kappa^2}{6\tau\lambda}\mathbb{E}[\|\mathbf{x}_t-\tilde{\mathbf{x}}_{t+1}\|^2]\right.\\
&+\gamma\sqrt{\delta}\left(-\frac{5\eta_t}{2\sqrt{\delta}}\mathbb{E}[\|\mathbf{w}_t-\nabla_{\mathbf{x}}f_\delta(\mathbf{x}_t,\mathbf{y}_t)\|^2]+\frac{m\eta_t^2\sigma_{\mathbf{x}}^2}{k^2}+\frac{4\sqrt{\delta}L_{fxy}^2\eta_t}{5}\mathbb{E}[\|\tilde{\mathbf{y}}_{t+1}-\mathbf{y}_t\|^2]+\frac{4\sqrt{\delta}c^2L^2d_1\eta_t}{5\delta^2}\mathbb{E}[\|\tilde{\mathbf{x}}_{t+1}-\mathbf{x}_t\|^2]\right)\\
&+\gamma\sqrt{\delta}\left(-\frac{125L_0^2}{6\sqrt{\delta}\tau^2}\eta_t\mathbb{E}[\|\mathbf{v}_t-\nabla_{\mathbf{y}}g(\mathbf{x}_t,\mathbf{y}_t)\|^2]+\frac{m\sigma_{\mathbf{y}}^2\eta_t^2}{k^2}+\frac{12\sqrt{\delta}\tau^2L_{fy}^2\eta_t}{125L_0^2}(\mathbb{E}[\|\tilde{\mathbf{x}}_{t+1}-\mathbf{x}_t\|^2]+\mathbb{E}[\|\tilde{\mathbf{y}}_{t+1}-\mathbf{y}_t\|^2])\right)\\
\leq&-\frac{\gamma\eta_t}{2}\mathbb{E}[\|\mathbf{w}_t-\nabla_{\mathbf{x}}f_\delta(\mathbf{x}_t,\mathbf{y}_t)\|^2]-\frac{L_0^2\gamma\eta_t}{4}\mathbb{E}[\|y_t-y^*(\mathbf{x}_t)\|^2]+\gamma\eta_tL_0^2\kappa^2\delta^2+\frac{\gamma m\sqrt{\delta}\eta_t^2}{k^2}(\sigma_{\mathbf{x}}^2+\frac{\sigma_{\mathbf{y}}^2}{M})\\
&-\eta_t\left[\frac{1}{2\gamma}-\frac{125^2L_0^2\gamma\kappa^2}{6\tau^2\lambda^2}-\frac{4\gamma c^2L^2d_1}{5\delta}-\frac{12\tau^2\gamma L_{fy}^2}{125L_0^2}\right]\mathbb{E}[\|\tilde{\mathbf{x}}_{t+1}-\mathbf{x}_t\|^2]\\
&-\eta_t\left[\frac{15L_0^2\gamma}{4\lambda\tau}-\frac{4\gamma L_{fxy}^2}{5}-\frac{12\tau^2\gamma L_{fy}^2}{125L_0^2}\right]\mathbb{E}[\|\tilde{\mathbf{y}}_{t+1}-\mathbf{y}_t\|^2]\\
\leq&-\frac{\gamma\eta_t}{4}\left(2\left[\left\|\mathbf{w}_t-\nabla_{\mathbf{x}}f_\delta(\mathbf{x}_t,\mathbf{y}_t)\right\|^2\right]+L_0^2\left[\left\|\mathbf{y}_t-\mathbf{y}^*(\mathbf{x}_t)\right\|^2\right]\right)-\frac{\eta_t}{4\gamma}\mathbb{E}\left[\left\|\tilde{\mathbf{x}}_{t+1}-\mathbf{x}_t\right\|^2\right]\\
&+\gamma\eta_tL_0^2\kappa^2\delta^2+\frac{m\gamma\sqrt{\delta}\eta_t^2}{k^2}\left(\frac{16\sqrt{2\pi}d_1L^2}{M}+\frac{\sigma_{\mathbf{y}}^2}{M}\right)
\end{aligned}\tag{87}$$

where the second inequality is due to $0<\delta<1$,the last inequality is due to $0<\gamma\leq\frac{1}{2\sqrt{\frac{125^2L_0^2\kappa^2}{6\tau^2\lambda^2}+\frac{4c^2L^2d_1}{5\delta}+\frac{12\tau^2L_{fy}^2}{125L_0^2}}}\leq\frac{1}{2\sqrt{\frac{125^2L_0^2\kappa^2}{6\tau^2\lambda^2}+\frac{4c^2L^2d_1}{5\delta}+\frac{12\tau^2L_{fy}^2}{125L_0^2}}}$ and $0<\lambda\leq\frac{15L_0^2}{4\tau\left(\frac{4L_{fxy}^2}{5}+\frac{12\tau^2L_{fy}^2}{125L_0^2}\right)}$ for all $t\geq 1$.

Let $\psi_t=2\left[\left\|\mathbf{w}_t-\nabla_{\mathbf{x}}f_\delta(\mathbf{x}_t,\mathbf{y}_t)\right\|^2\right]+L_0^2\left[\left\|\mathbf{y}_t-\mathbf{y}^*(\mathbf{x}_t)\right\|^2\right]+L_0^2\kappa^2\delta^2$, we have

$$\frac{\gamma\eta_t}{4}\psi_t+\frac{\eta_t}{4\gamma}\mathbb{E}\left[\left\|\tilde{\mathbf{x}}_{t+1}-\mathbf{x}_t\right\|^2\right]\leq\Gamma_t-\Gamma_{t+1}+\frac{5\gamma\eta_tL_0^2\kappa^2\delta^2}{4}+\frac{m\gamma\sqrt{\delta}\eta_t^2}{k^2}\left(\frac{16\sqrt{2\pi}d_1L^2}{M}+\frac{\sigma_{\mathbf{y}}^2}{M}\right)\tag{88}$$

Taking average over $t=1,\cdots,T$ on both sides, we have

$$\frac{1}{T}\sum_{t=1}^T\mathbb{E}\left[\frac{\eta_t}{4}\psi_t+\frac{\eta_t}{4\gamma^2}\left\|\tilde{\mathbf{x}}_{t+1}-\mathbf{x}_t\right\|^2\right]\leq\sum_{t=1}^T\frac{(\Gamma_t-\Gamma_{t+1})}{T\gamma}+\frac{5}{4T}\sum_{t=1}^T\eta_tL_0^2\kappa^2\delta^2+\frac{1}{T}\sum_{t=1}^T\frac{m\sqrt{\delta}\eta_t^2}{k^2}\left(\frac{16\sqrt{2\pi}d_1L^2}{M}+\frac{\sigma_{\mathbf{y}}^2}{M}\right)\tag{89}$$

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Define $\Delta_1 = \|\mathbf{y}_1 - \mathbf{y}^*(x_1)\|^2$, we have

$$\begin{aligned}\Gamma_1 =&\mathbb{E}[\Phi(\mathbf{x}_1) + \frac{5L_0^2\gamma}{\lambda\tau}\|\mathbf{y}_1 - \mathbf{y}^*(\mathbf{x}_1)\|^2 + \gamma\sqrt{\delta}(\|\mathbf{v} - \nabla_{\mathbf{y}} g(\mathbf{x}_1, \mathbf{y}_1)\|^2 + \|\mathbf{w}_1 - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_1, \mathbf{y}_1)\|^2)] \\ =&\Phi(\mathbf{x}_1) + \frac{5L_0^2\gamma}{\lambda\tau}\Delta_1 + \sqrt{\delta}\gamma\left(\frac{16\sqrt{2\pi}d_1L^2}{M} + \frac{\sigma_{\mathbf{y}}^2}{M}\right) \end{aligned} \tag{90}$$

Since $\eta_t = \frac{k}{(m+t)^{1/2}}$, we have $\eta_T^{-1} \geq \eta_t^{-1}$ and

$$\begin{aligned}&\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left[\frac{1}{4}\psi_t + \frac{1}{4\gamma^2}\left\|\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\right\|^2\right] \\ \leq&\sum_{t=1}^{T}\frac{(\Gamma_t - \Gamma_{t+1})}{T\eta_T\gamma} + \frac{5}{4T\eta_T}\sum_{t=1}^{T}\eta_t L_0^2\kappa^2\delta^2 + \frac{1}{T\eta_T}\sum_{t=1}^{T}\frac{m\sqrt{\delta}\eta_t^2}{k^2}\left(\frac{16\sqrt{2\pi}d_1L^2}{M} + \frac{\sigma_{\mathbf{y}}^2}{M}\right) \\ \leq&\sum_{t=1}^{T}\frac{(\Gamma_t - \Gamma_{t+1})}{T\eta_T\gamma} + \frac{5L_0^2\kappa^2\delta^2}{4T\eta_T}\int\frac{k}{(m+t)^{1/2}}dt + \frac{m\sqrt{\delta}\left(16\sqrt{2\pi}d_1L^2 + \sigma_{\mathbf{y}}^2\right)}{MT\eta_T k^2}\int\frac{k^2}{m+t}dt \\ \leq&\frac{(\Gamma_1 - \Phi^*)}{T\eta_T\gamma} + \frac{5L_0^2\kappa^2\delta^2 k}{4T\eta_T}(m+T)^{1/2} + \frac{m\sqrt{\delta}\left(16\sqrt{2\pi}d_1L^2 + \sigma_{\mathbf{y}}^2\right)}{MT\eta_T}\ln(m+T) \\ \leq&\left(\frac{(\Gamma_1 - \Phi^*)}{k\gamma} + \frac{m\sqrt{\delta}\left(16\sqrt{2\pi}d_1L^2 + \sigma_{\mathbf{y}}^2\right)}{Mk}\ln(m+T)\right)\frac{(m+T)^{1/2}}{T} + \frac{5L_0^2\kappa^2\delta^2}{4T}(m+T) \\ =&\left(\frac{(\Phi(\mathbf{x}_1) - \Phi^*)}{k\gamma} + \frac{5L_0^2}{\lambda\tau k}\Delta_1 + \frac{\sqrt{\delta}\left(16\sqrt{2\pi}d_1L^2 + \sigma_{\mathbf{y}}^2\right)}{Mk} + \frac{m\sqrt{\delta}\left(16\sqrt{2\pi}d_1L^2 + \sigma_{\mathbf{y}}^2\right)}{Mk}\ln(m+T)\right)\frac{(m+T)^{1/2}}{T} \\ &+ \frac{5L_0^2\kappa^2\delta^2}{4T}(m+T)\end{aligned} \tag{91}$$

Therefore, let $G = \frac{(\Phi(\mathbf{x}_1) - \Phi^*)}{k\gamma} + \frac{5L_0^2}{\lambda\tau k}\Delta_1 + \frac{\sqrt{\delta}\left(16\sqrt{2\pi}d_1L^2 + \sigma_{\mathbf{y}}^2\right)}{Mk} + \frac{m\sqrt{\delta}\left(16\sqrt{2\pi}d_1L^2 + \sigma_{\mathbf{y}}^2\right)}{Mk}\ln(m+T)$, we have

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left[\frac{1}{4}\psi_t + \frac{1}{4\gamma^2}\left\|\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\right\|^2\right] \leq \frac{G}{T}(m+T)^{1/2} + \frac{5L_0^2\kappa^2\delta^2}{4T}(m+T) \tag{92}$$

We have

$$\begin{aligned}\mathcal{M}_t =&\frac{1}{\gamma}\|\mathbf{x}_t - \tilde{\mathbf{x}}_{t+1}\| + (L_0\|\mathbf{y}^*(\mathbf{x}_t) - \mathbf{y}_t\| + \sqrt{2}\|\nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t) - \mathbf{w}_t\| + L_0\kappa\delta) \\ \geq&\frac{1}{\gamma}\|\mathbf{x}_t - \tilde{\mathbf{x}}_{t+1}\| + \|\nabla\Phi(\mathbf{x}_t) - \mathbf{w}_t\| \\ =&\|\mathbf{w}_t\| + \|\nabla\Phi(\mathbf{x}_t) - \mathbf{w}_t\| \\ \geq&\|\nabla\Phi(\mathbf{x}_t)\|\end{aligned} \tag{93}$$

where the first equality is due to $\tilde{\mathbf{x}}_{t+1} = \mathbf{x}_t - \gamma\mathbf{w}_t$. Then, we have

$$\|\nabla\Phi(\mathbf{x}_t)\| \leq \|\mathcal{M}_t\| \tag{94}$$

According to the Cauchy-Schwarz inequality, we have

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\|\nabla\Phi(\mathbf{x}_t)\|] \leq \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\mathcal{M}_t] \leq \sqrt{\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\mathcal{M}_t^2]} \tag{95}$$

Then, we have

$$\begin{aligned}\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\mathcal{M}_t^2] \leq&\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[4L_0^2\|\mathbf{y}_t - \mathbf{y}^*(\mathbf{x}_t)\|^2 + 8\|\mathbf{w}_t - \nabla_{\mathbf{x}} f_\delta(\mathbf{x}_t, \mathbf{y}_t)\|^2 + 4L_0^2\kappa^2\delta^2 + \frac{4\|\tilde{\mathbf{x}}_{t+1} - \mathbf{x}_t\|^2}{\gamma^2}] \\ \leq&\frac{16G}{T}(m+T)^{1/2} + \frac{20L_0^2\kappa^2\delta^2}{T}(m+T)\end{aligned} \tag{96}$$

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Thus, we have

$$\sqrt{\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\mathcal{M}_t^2]} \leq \frac{4\sqrt{G}}{T^{1/2}}(m+T)^{1/4} + \frac{2\sqrt{5}L_0\kappa\delta}{T^{1/2}}(m+T)^{1/2}$$
$$\leq \frac{4\sqrt{G}m^{1/4}}{T^{1/2}} + \frac{4\sqrt{G}}{T^{1/4}} + \frac{2\sqrt{5m}L_0\kappa\delta}{T^{1/2}} + 2\sqrt{5}L_0\kappa\delta \quad (97)$$

Therefore, we have

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\|\nabla\Phi(\mathbf{x}_t)\|] \leq \sqrt{\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\mathcal{M}_t^2]} \leq \frac{4\sqrt{G}m^{1/4}}{T^{1/2}} + \frac{4\sqrt{G}}{T^{1/4}} + \frac{2\sqrt{5m}L_0\kappa\delta}{T^{1/2}} + 2\sqrt{5}L_0\kappa\delta \quad (98)$$

Let $m \geq \frac{8\ell^2k^2}{\frac{1251^2L_0^2\kappa^2}{6\tau^2\lambda^2}+\frac{4c^2L^2d_1}{5\delta}+\frac{12\tau^2L^2f_y}{125L_0^2}}$. Without loss of generality, it is easy to verify that $k = \mathcal{O}(1)$, $\ell = \mathcal{O}(d_1^{1/2}\delta^{-1})$, $m = \mathcal{O}(\delta^{-1})$, $\lambda^{-1} = \mathcal{O}(\tau)$, $\gamma^{-1} = \mathcal{O}((d_1/\delta)^{1/2})$, $G = \mathcal{O}((d_1/\delta)^{1/2} + d_1M^{-1}\delta^{-1/2}ln(m+T))$, $\delta = \mathcal{O}(\epsilon)$. Then, choosing $M = \mathcal{O}(\sqrt{d_1})$, we have $G = \tilde{\mathcal{O}}((d_1/\delta)^{1/2})$. Then, under the condition $\mathbb{E}[\|\nabla\Phi(\mathbf{x})\|] \leq \epsilon$ and $M = \mathcal{O}(\sqrt{d_1})$, our Algorithm 1 has a query complexity of $\tilde{\mathcal{O}}(d_1^{1.5}\delta^{-1}\epsilon^{-4}\})$ for finding a (δ,ϵ)-GGSP defined in Definition 4.1. This complexity result is consistent with the known complexity of zeroth-order methods for nonsmooth minimization [18, 22]. □

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